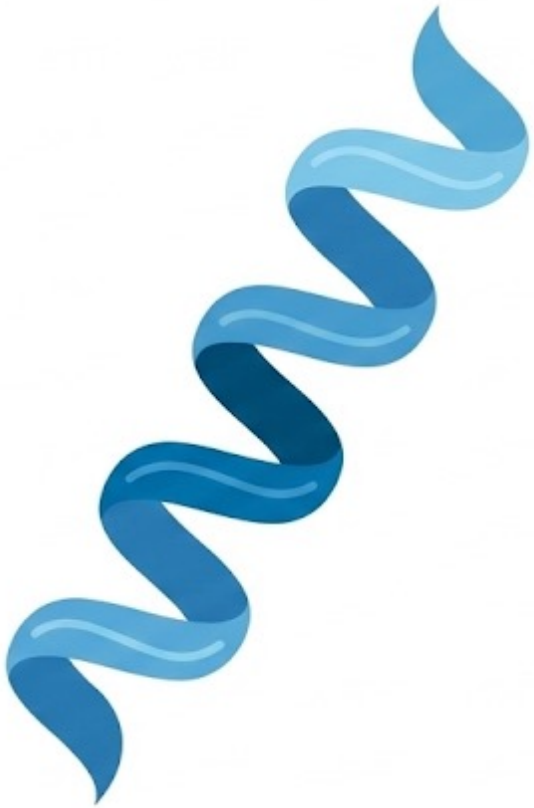
Biology 3030: Biochemistry

Spring 2026

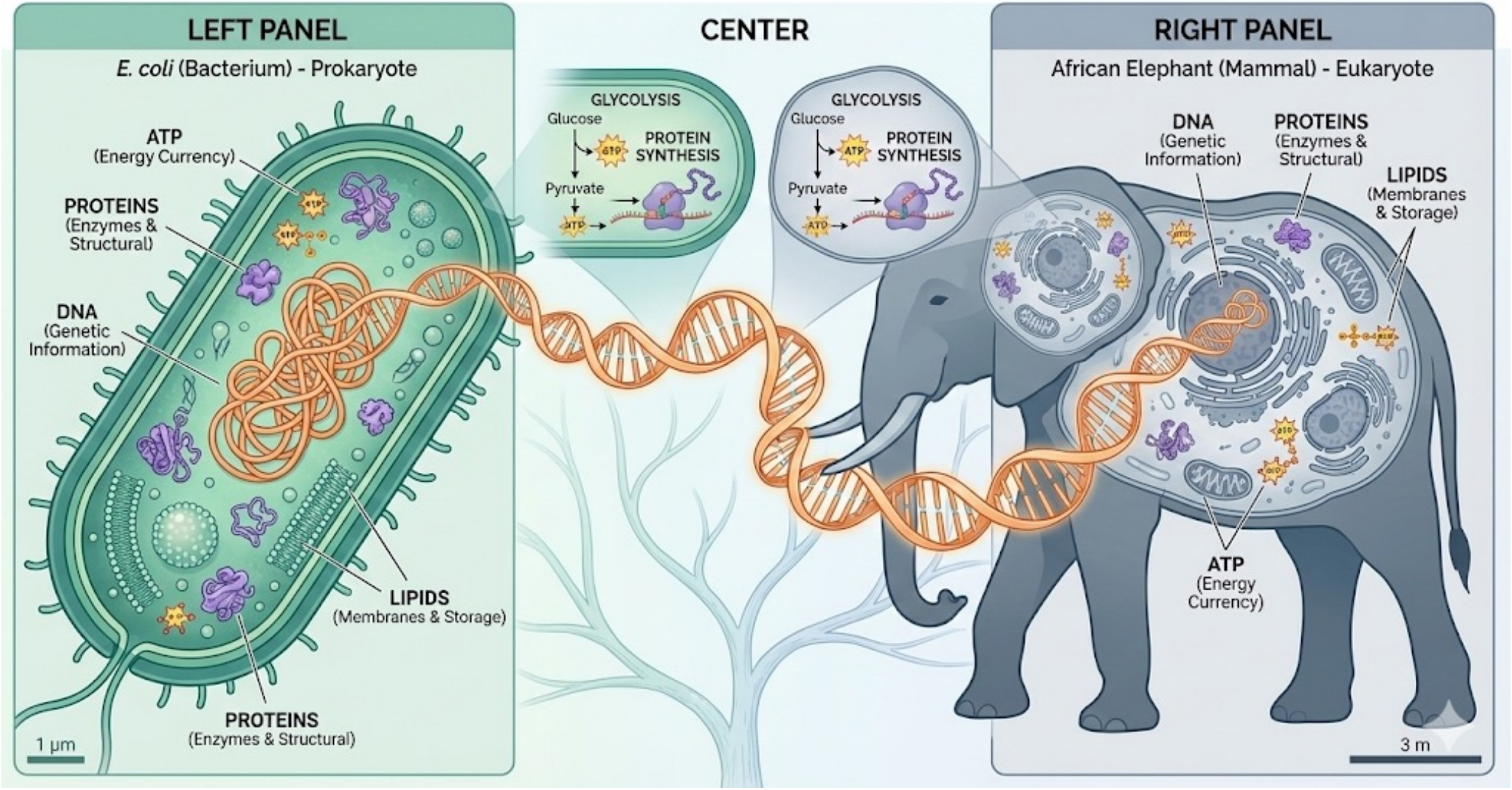


Today's Goals

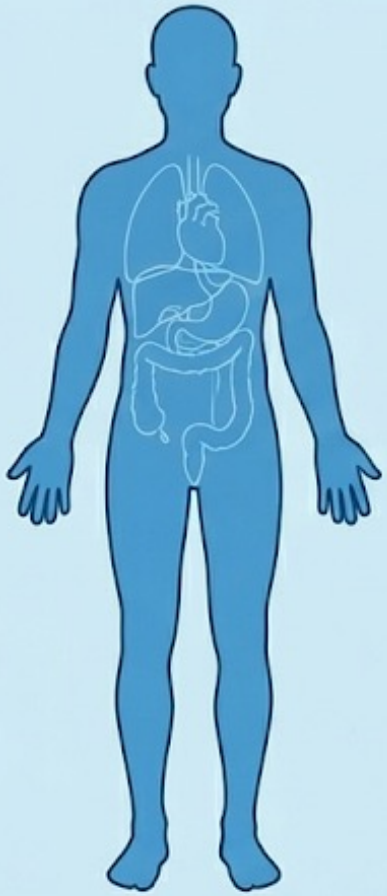
- Why biochemistry matters (and why now)
- How this course works
- How to succeed
- Getting started
- [BREAK]
- Chapter 1: The Unity of Life

Why We're Here

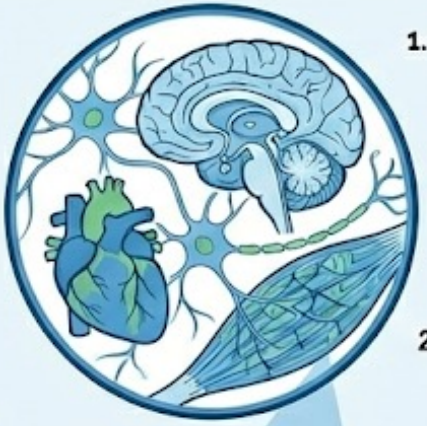
‘BIOCHEMICAL UNITY ACROSS LIFE: FROM BACTERIA TO MAMMALS’



THE MOLECULAR BASIS OF LIFE

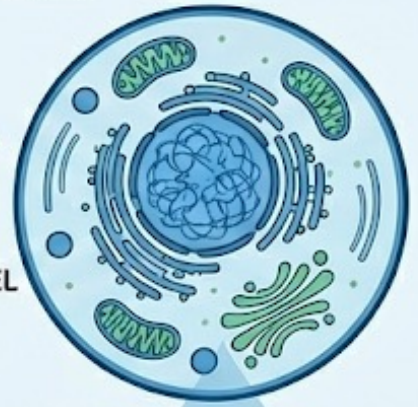


MACROSCOPIC TO MICROSCOPIC



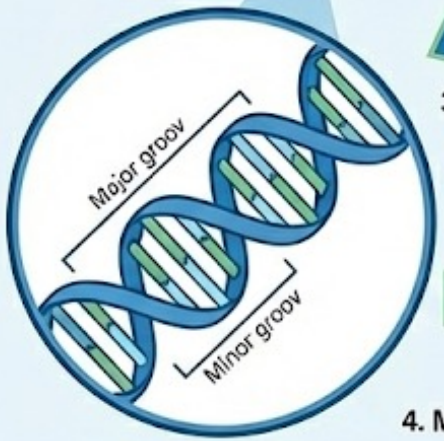
1. ORGAN & TISSUE LEVEL

ZOOM IN



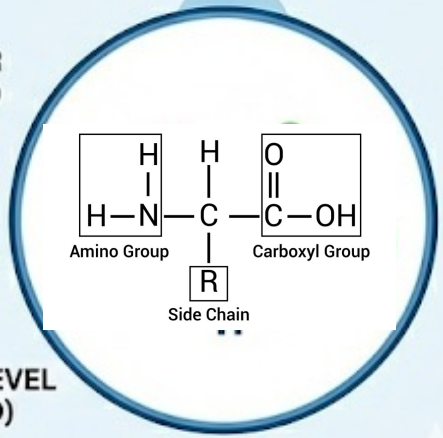
2. CELLULAR LEVEL

ZOOM IN



3. MOLECULAR LEVEL (DNA)

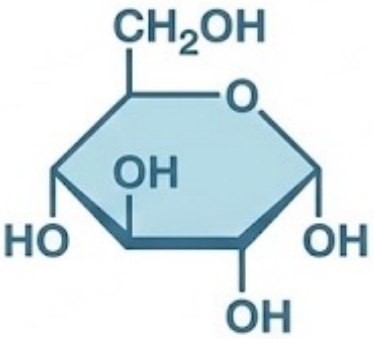
ZOOM IN



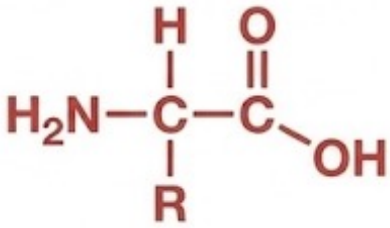
4. MOLECULAR LEVEL (AMINO ACID)

Everything you experience is mediated by molecules. Understanding this language reveals the rules of biochemistry.

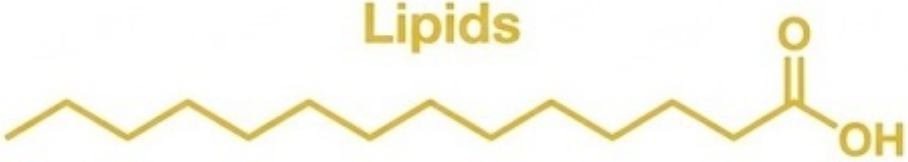
Carbohydrates



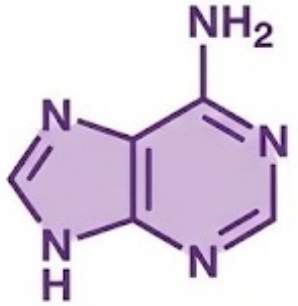
Proteins



Lipids



Nucleic Acids

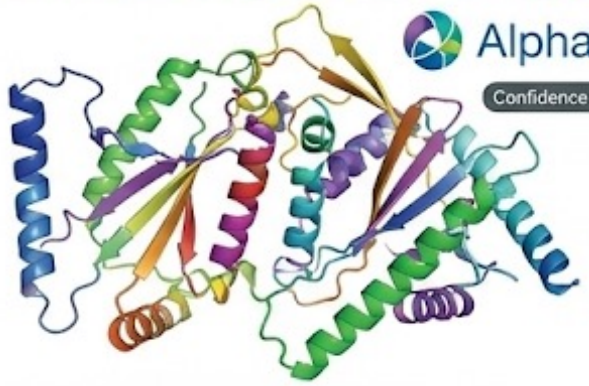


The Future is Biochemistry

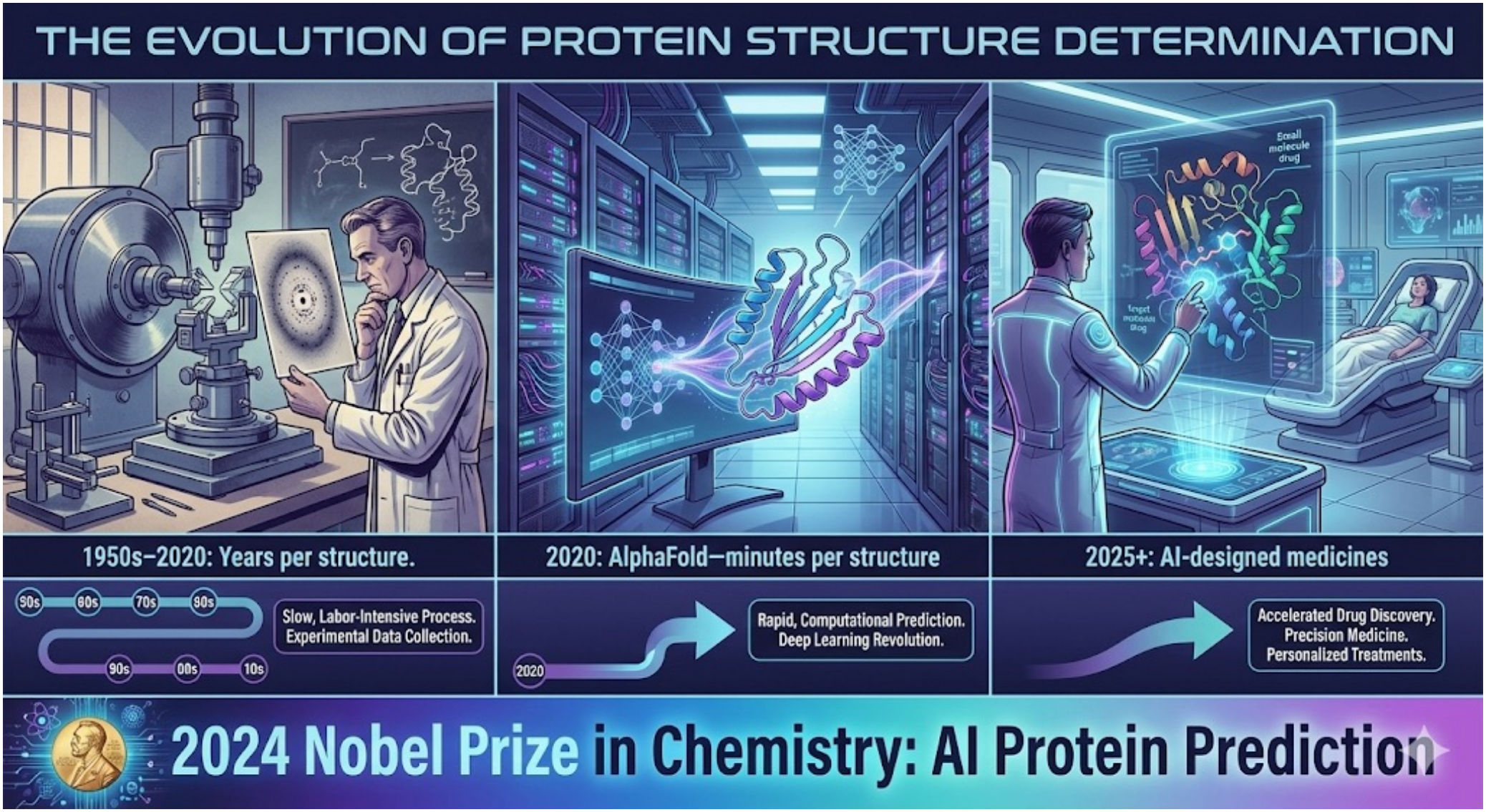
GENOME EDITING (CRISPR)

ADVANCED VACCINES

PROTEIN DESIGN (AlphaFold)

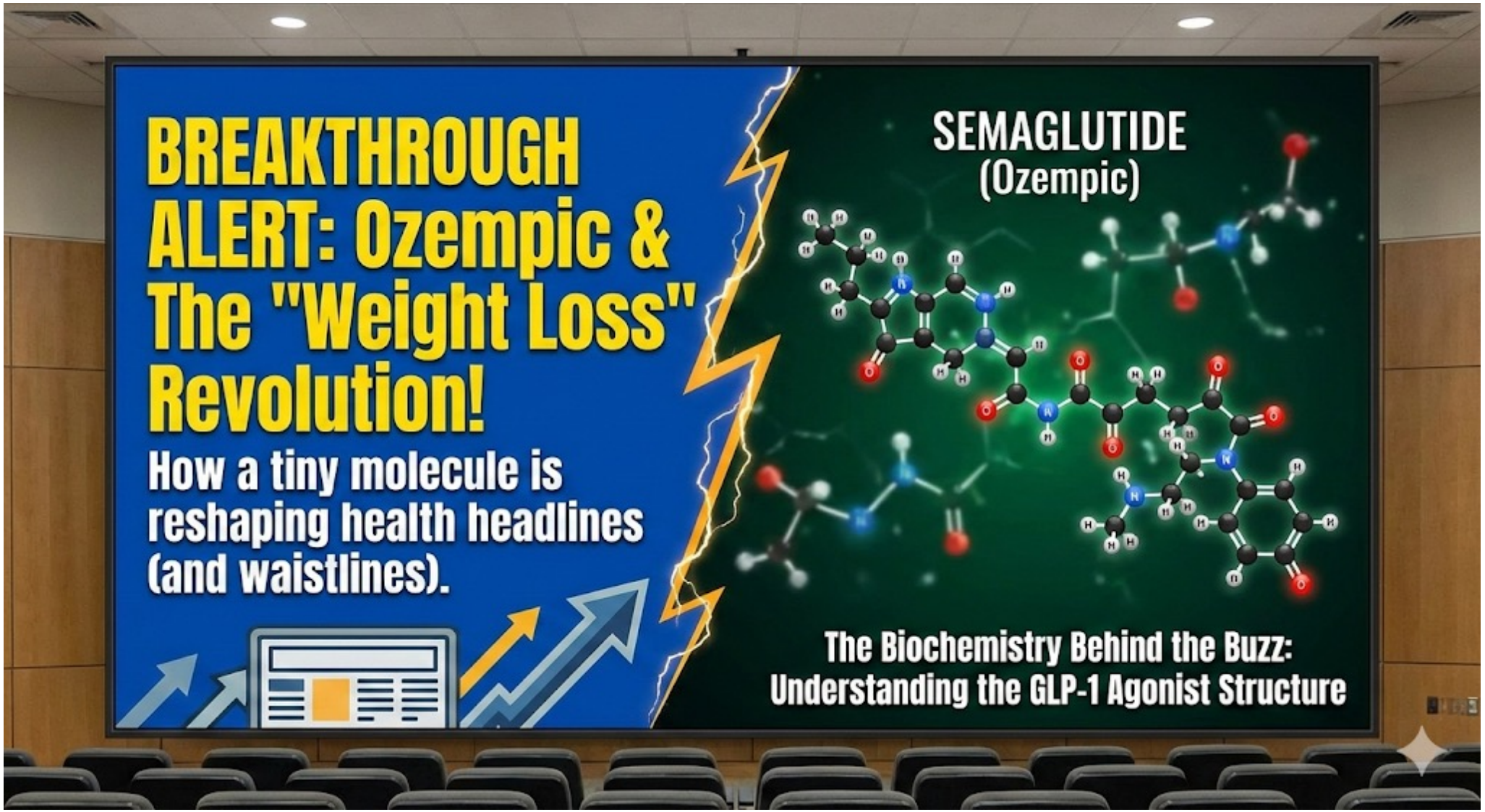


DRUG DISCOVERY



- AlphaFold turned protein structure prediction into something fast and scalable
- Why you should care: structure helps explain function, disease, and drug design
- This course teaches you the rules underneath the predictions

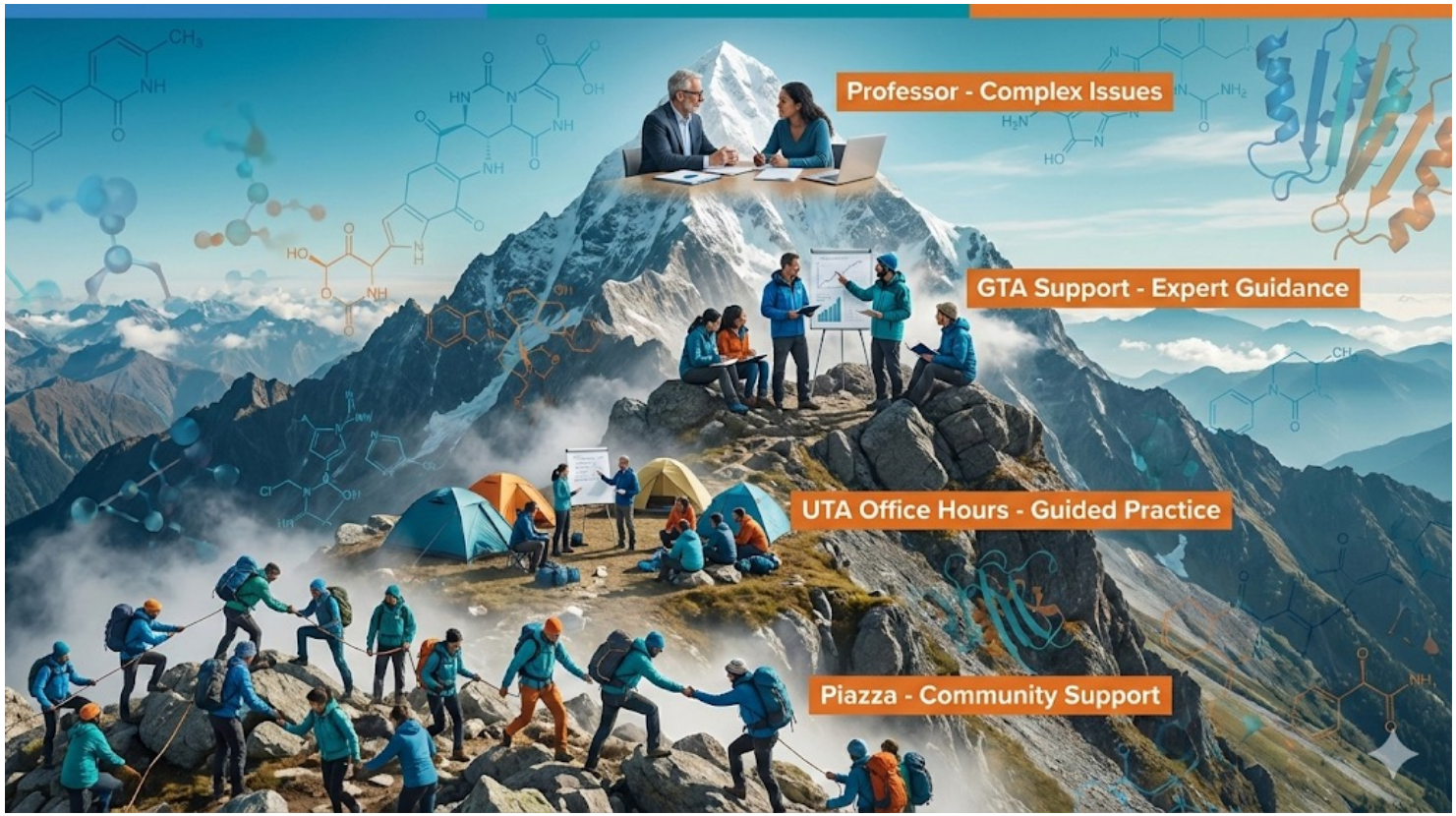
Why do we care?



SCIENCE
SCIENCE
CH3
N
Medicine's Secret Weapon

This will be challenging

Edmund Hilary (you) Tenzig Norgay (me)



How This Course Works

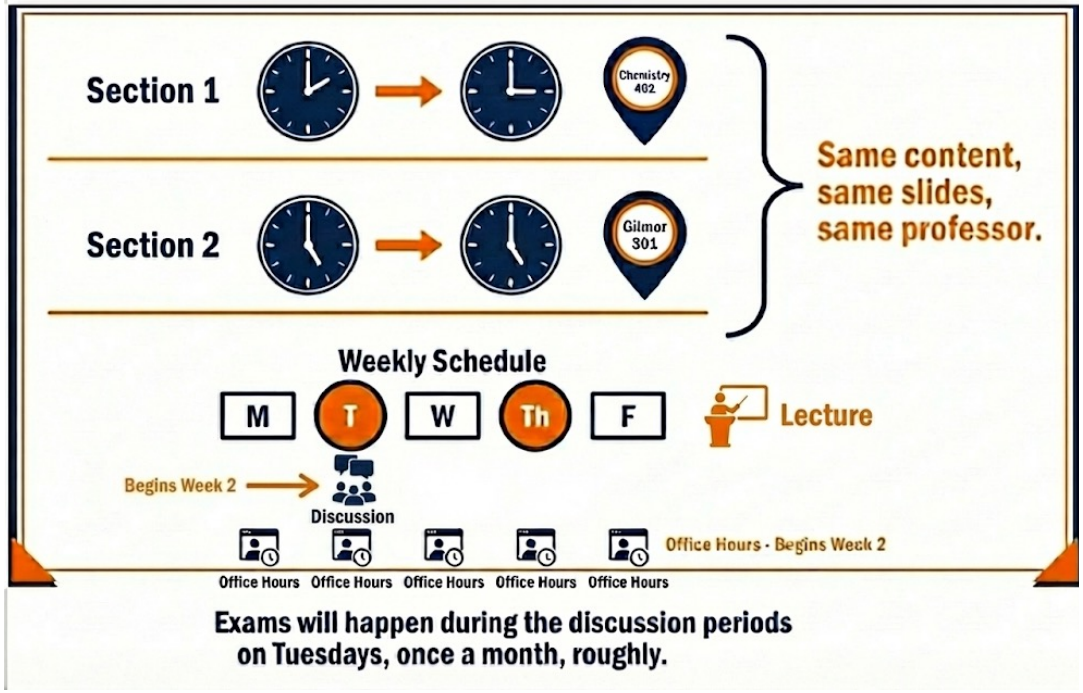
BIOL 3030: Biochemistry

Spring 2026

Credits: Three credit hours
Lecture: 2 sections: Tuesday/Thursday 2:00-3:15pm and 3:30-4:45 Chemistry 402
Discussion: 2 sections: Tuesday 5:00-7:00pm, Chemistry 402 and Gilmer 301

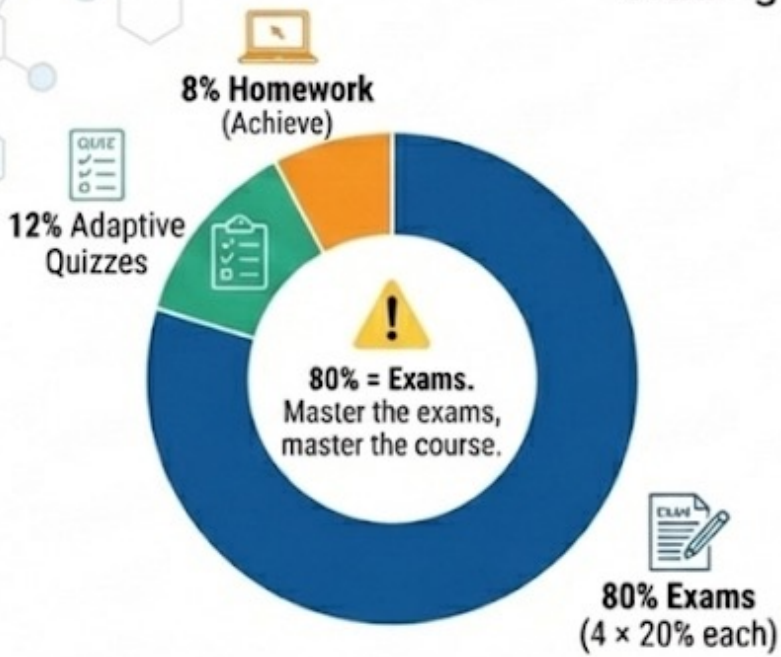
Instructor

Christopher Deppmann
Email: deppmann@virginia.edu
Office Hours: Monday 11:00am-12:00pm
Location: https://virginia.zoom.us/j/2854285879
For individual appointments at other times, please email to schedule.

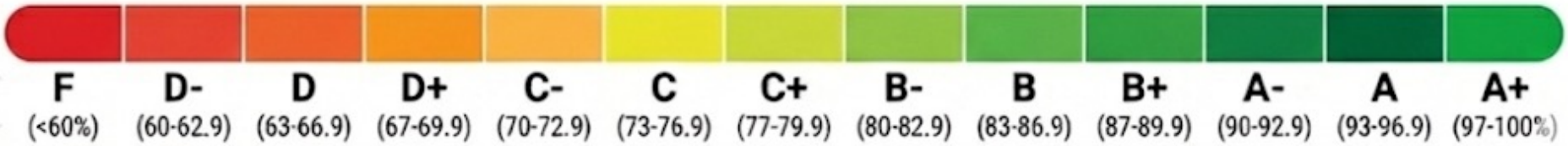


Logistics

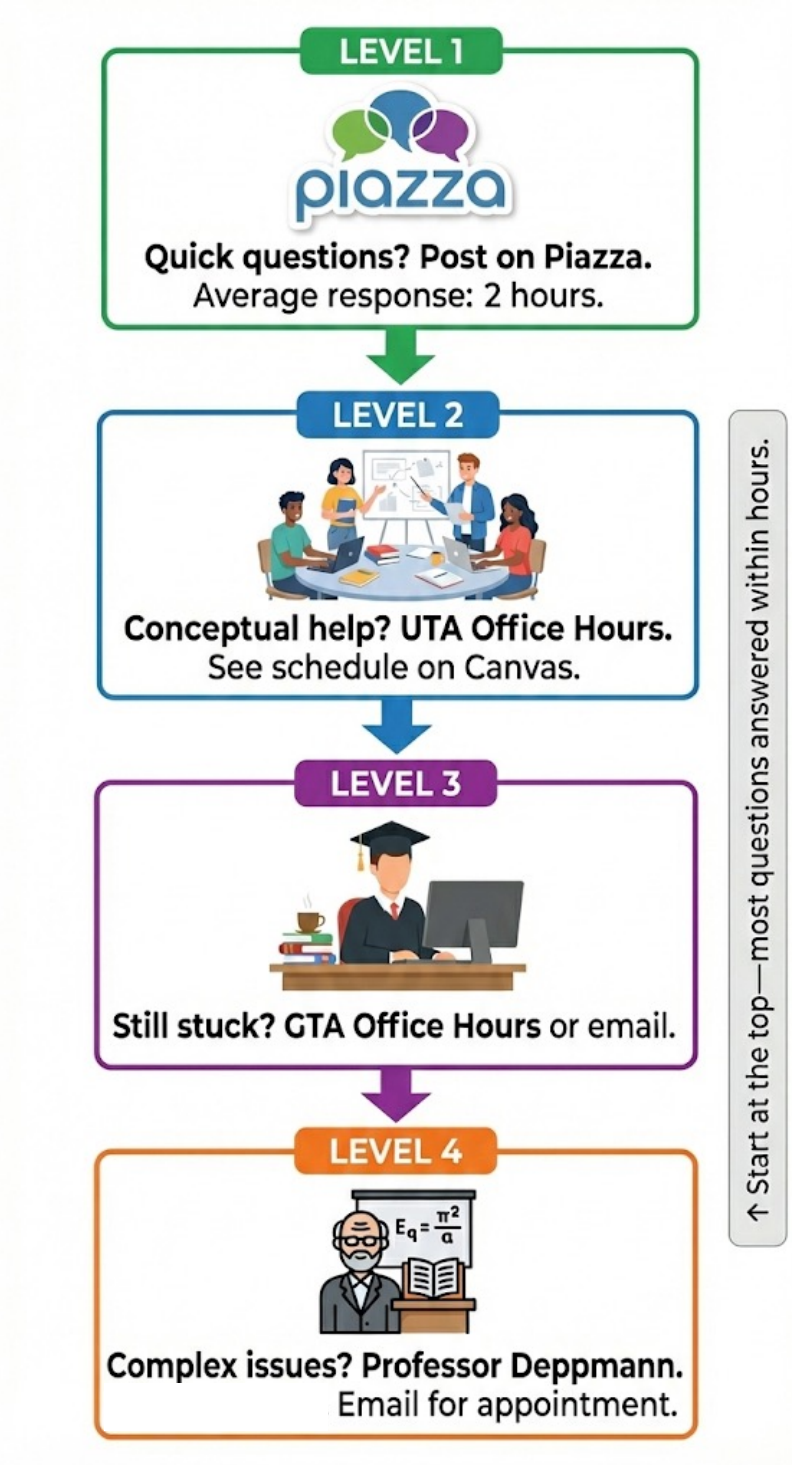
Grading Structure (Spring 2026)



Component	Weight	Details	Drop Policy
Exams	80%	4 exams at 20% each	Drop lowest (best 3 at 26.67% each)
Adaptive Quizzes	12%	~30 quizzes, graded for correctness	Drop 2 lowest
Homework	8%	~30 assignments, graded for correctness	Drop 2 lowest
Extra Credit	Up to 2-3 pts/exam	Announced in discussion sessions	Added as percentage points to exam scores



Getting help: Instructional team



Graduate TAs

- **Jennie Nguyen:** war2ca@virginia.edu
- **Monika Wieliniec:** qdc5xu@virginia.edu
- **Megan Garris:** fmk3uc@virginia.edu
- **Andrew Spira:** abs6ken@virginia.edu

Undergrad TAs and Question Escalation

Last Name Prefix Range	Pod	Assigned UTA	UTA email	Assigned GTA	GTA email
AA - BA	1	Emma Guo	tbg2rr@virginia.edu	Andrew Spira	abs6ken@virginia.edu
BE - CHEN	2	Sarah Cheon	ucy5db@virginia.edu	Andrew Spira	abs6ken@virginia.edu
CHH - DR	3	Sebastian Moeller	hpp3nu@virginia.edu	Andrew Spira	abs6ken@virginia.edu
DU - GHO	4	Annie Zhao	ttd8us@virginia.edu	Andrew Spira	abs6ken@virginia.edu
GI - HOO	5	Abhi Kodali	mkp7bd@virginia.edu	Jennie Nguyen	war2ca@virginia.edu
HOU-KHA	6	Julia Kim	uwq6qw@virginia.edu	Jennie Nguyen	war2ca@virginia.edu
KI - LEE	7	Alina Alfaro	sgw3ms@virginia.edu	Jennie Nguyen	war2ca@virginia.edu
LEI - MEE	8	Phillip Friedman	vds6vh@virginia.edu	Jennie Nguyen	war2ca@virginia.edu
MEH - NGU	9	Tanmayi Alaparti	qbd7vb@virginia.edu	Megan Garris	fmk3uc@virginia.edu
NI - PRA	10	Nasim Abib	rhu2xr@virginia.edu	Megan Garris	fmk3uc@virginia.edu
PRE - SCA	11	Hemanya Bhatt	jyb3bt@virginia.edu	Megan Garris	fmk3uc@virginia.edu
SCH - STR	12	Deetya Gupta	zpz3vw@virginia.edu	Monika Wieliniec	qdc5xu@virginia.edu
SUH - WAN	13	Maggie Flynn	rgr5eg@virginia.edu	Monika Wieliniec	qdc5xu@virginia.edu
WAT - ZHO	14	Autumn Bisset	wnp7uy@virginia.edu	Monika Wieliniec	qdc5xu@virginia.edu

Office Hours

	Monday	Tuesday	Wednesday	Thursday	Friday
9:00		9-10 am OFFICE HOURS		9-10 am OFFICE HOURS	
9:15					
9:30					
9:45					
10:00					10 - 11 am OFFICE HOURS
10:15					
10:30					
10:45					
11:00	11am - 12 pm DR. DEPPMANN'S OFFICE HOURS				
11:15					
11:30					
11:45					
12:00			12 - 1 pm OFFICE HOURS		
12:15					
12:30		12:30 - 1:30 pm OFFICE HOURS		12:30 - 1:30 pm OFFICE HOURS	
12:45					
13:00					1 - 2 pm OFFICE HOURS
13:15					
13:30					
13:45					
14:00					
14:15		Lectures		Lectures	
14:30					
14:45					
15:00					
15:15	3:00 - 4:00 pm OFFICE HOURS				
15:30					
15:45					
16:00					
16:15					
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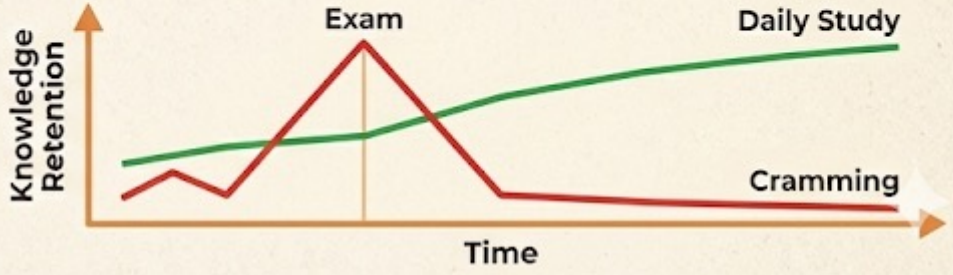
16:45					
17:00		Disc.			
17:15					
17:30					
17:45					
18:00					
18:15	6 - 7 pm OFFICE HOURS		6 - 7 pm OFFICE HOURS		
18:30					

Zoom link for office hours-(note that this is different from Deppmann's)
https://virginia.zoom.us/j/94996349116?pwd=sF0tCAJYpcaCj93aqLBfzFBNacYap1.1

6-hour cram session before exam

30 minutes daily



Biochemistry Course: Assignments & Success Strategies

QUIZZES = STAY ON TRACK

Due the day of lecture

Designed for active learning and feedback

Keep answering until you reach full credit

No reason not to earn 100 percent

Early Semester Grace Period

First few quizzes

No late penalty for being a day or two late

After Grace Period

Late penalty is 10 points per day

Stop early and the score at that point is what gets recorded.

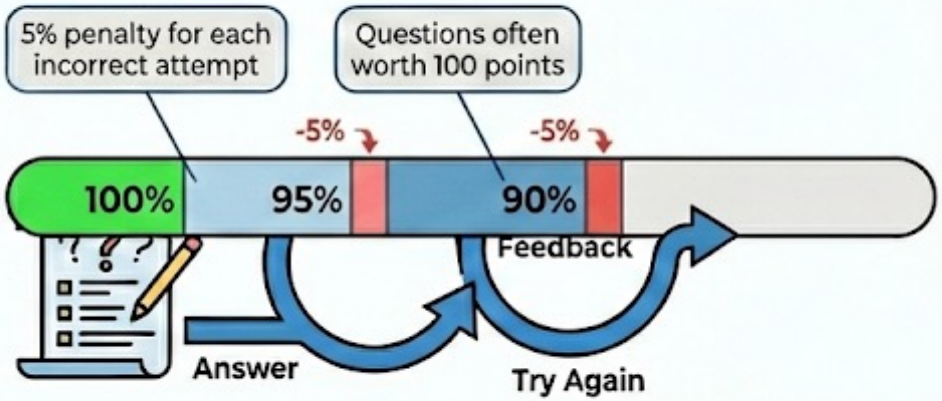
HOMEWORK = EXAM PREP

Due the day before each exam

No extensions available

Purpose: Help you prepare for the exam

Unlimited attempts per question

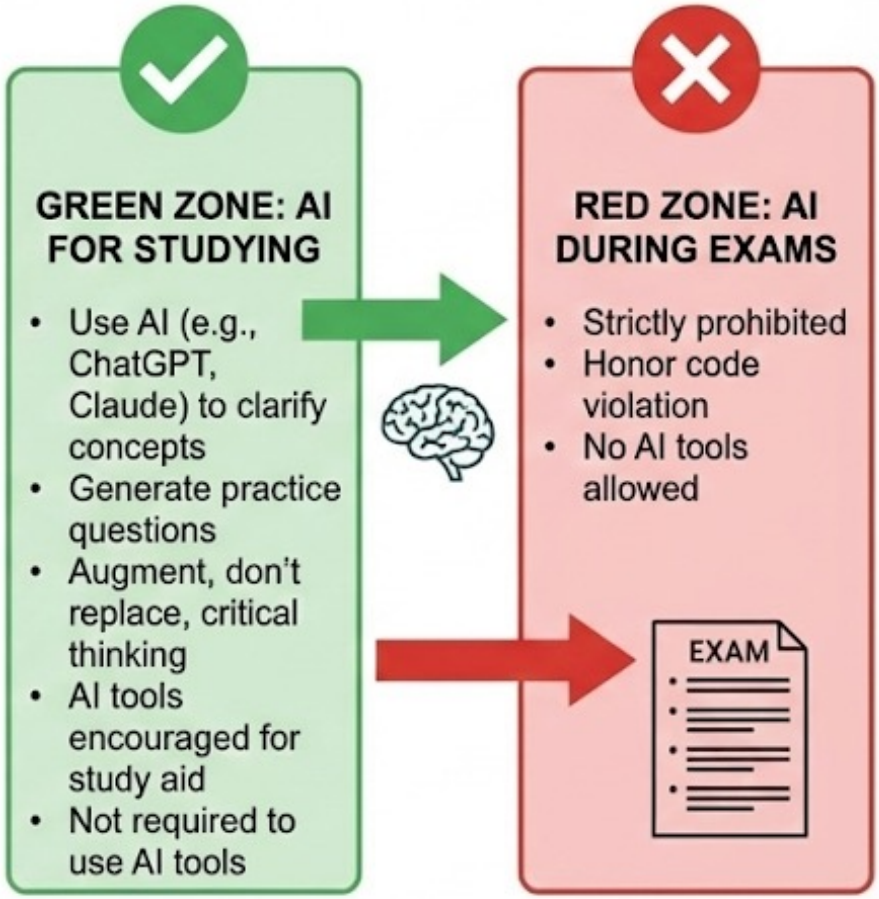


These assignments reward persistence, not perfection

AI in Our Biochemistry Class

AI tools (e.g., ChatGPT, Claude) encouraged for study aid

- Responsibility for understanding material remains yours
- I use AI to enhance course prep and content
- AI integration makes course more dynamic and relevant
- Learn to work effectively with AI
- it's here to stay!
- If uncomfortable, course fit



Student Guidelines:

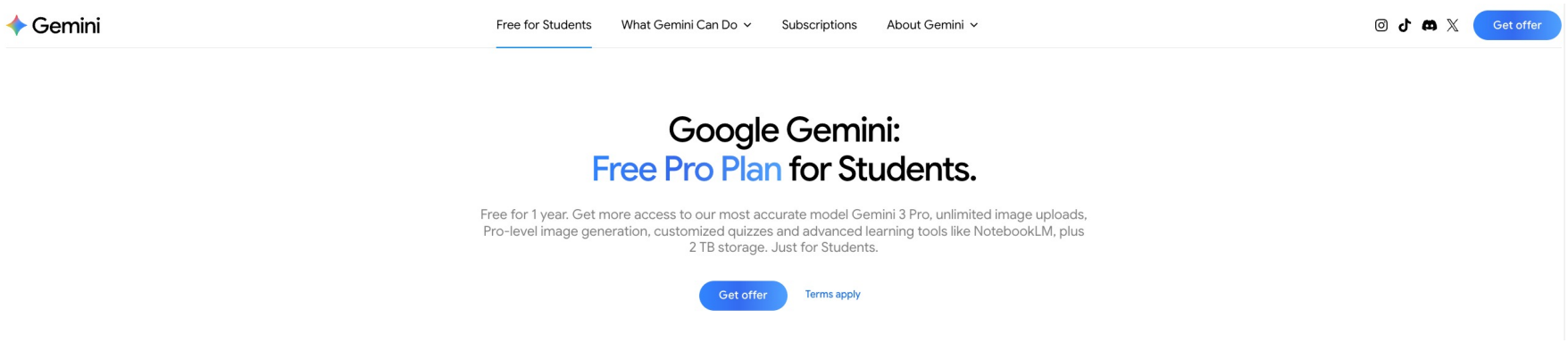
- Use AI to clarify concepts and generate practice questions
- AI should augment, not replace, your critical thinking
- You are not required to use AI tools

Remember:

- AI has limitations and potential biases
- Developing AI literacy is crucial for your future careers

Remember:
AI for studying = YES.
AI during exams = HONOR CODE VIOLATION.

https://gemini.google/students/



Demo of Linus

https://chatgpt.com/g/g-AbHbD9UtV-linus-gpt

Exam Schedule:

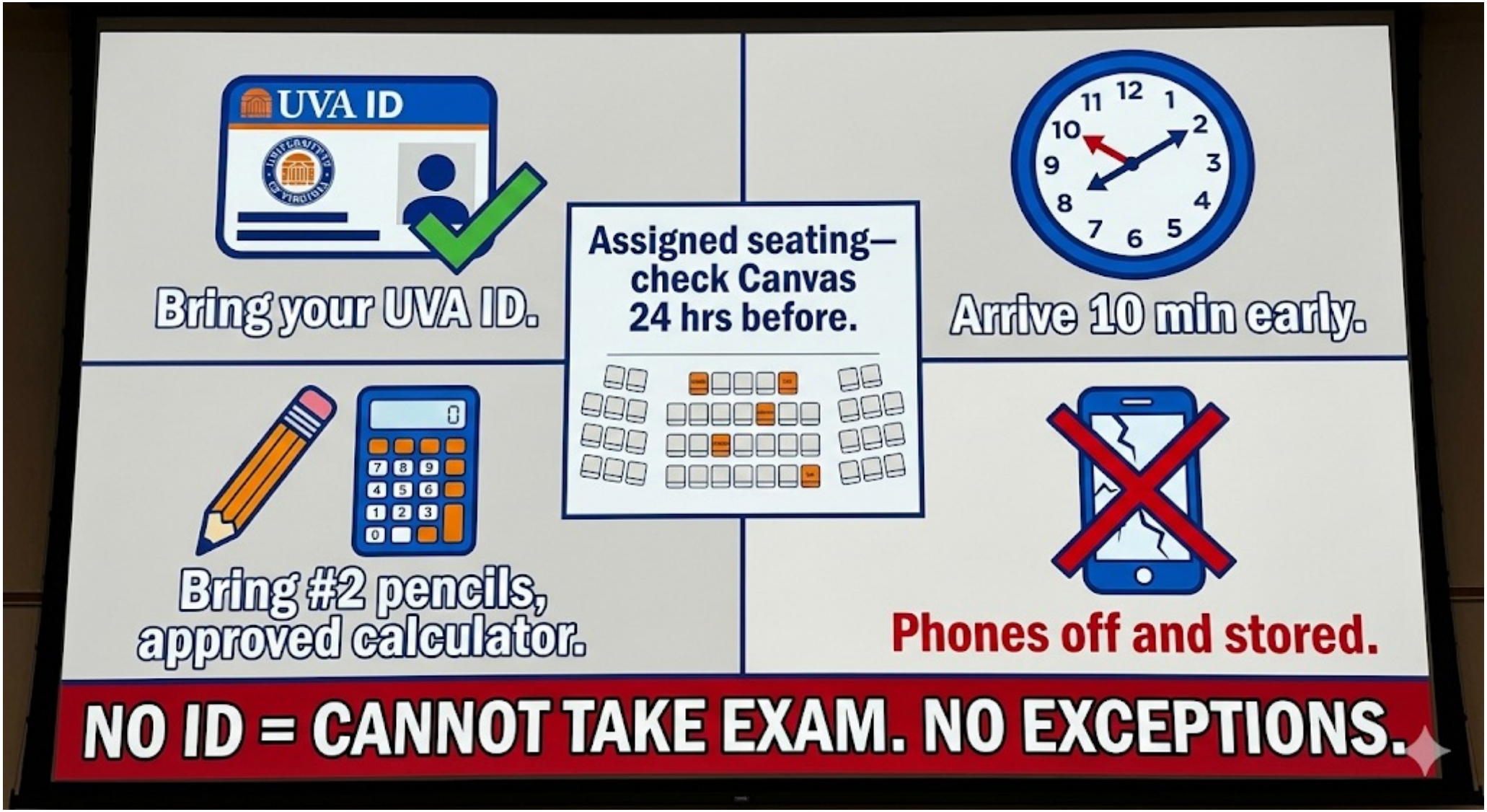
Exam	Date	Coverage
Exam 1	Tuesday, February 10, 2026	Chapters 1-8
Exam 2	Tuesday, March 17, 2026	Chapters 9-15
Exam 3	Tuesday, April 7, 2026	Chapters 16-22
Exam 4	Finals Week (Date TBA)	Chapters 23-30

Exam Format:

- • Closed-book, multiple choice format
- • 2 hours to complete
- • Multiple exam versions to maintain integrity
- • Simple 4-function calculators permitted (no phones or programmable calculators)
- • Must present valid UVA student ID to receive and submit exam

Exams

Four exams will be given during the semester to assess your learning and ability to integrate concepts. Exams will be administered during the Tuesday discussion time (5:00-7:00pm) except for the final exam, which will be during finals week.


UVA ID
Bring your UVA ID.
Assigned seating—
check Canvas
24 hrs before.
Arrive 10 min early.
Bring #2 pencils,
approved calculator.
Phones off and stored.
NO ID = CANNOT TAKE EXAM. NO EXCEPTIONS.

Sunday Study Sessions
When: Sunday before each exam, 3:00 PM
Where: [TBA - announced on Canvas] - ALSO ON ZOOM
What: Group review, practice problems, pathway drawing
Bring: Questions, notes, study materials
zoom

Extra Credit

Yo, I'm diving in that substrate mix, ah
Reaction speed, it gets no quicker, nah
Active site, lock and key
Drop the pH, watch it flee, ah
Turnover, check the rate
Michaelis-Menten state
I'm the enzyme in your scene
Breaking bonds like I'm supreme, yeah
Pump the product, keep it clean
[Verse: Kinetic King]
(Yeah, let's go, let's go)
When I'm catalyzing, it's like a blitz, ah
Kinetics sharp like a catalyst whiz, cha
Saturation high, Vmax we hit
Km so low, competitors quit, yeah
Affinity tight, substrate's the bait
Enzyme's delight, complexity's fate
Inhibition? Naw, I just innovate
My active realm, where I dominate, aye

Podcast

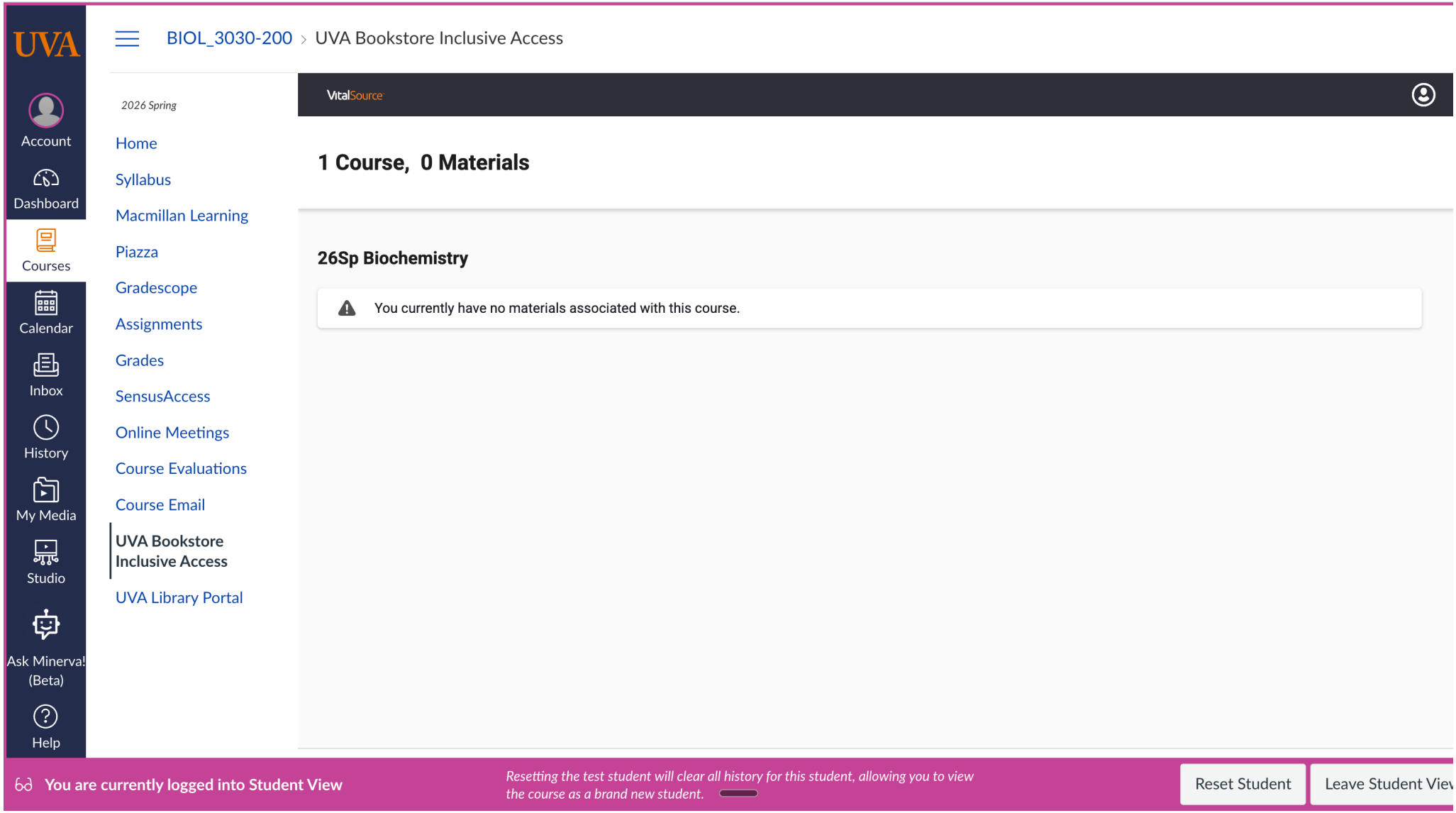
https://open.spotify.com/show/6gUAeZTYYRPzdAc84Lj4Gt?si=d8fb877e6adc4393

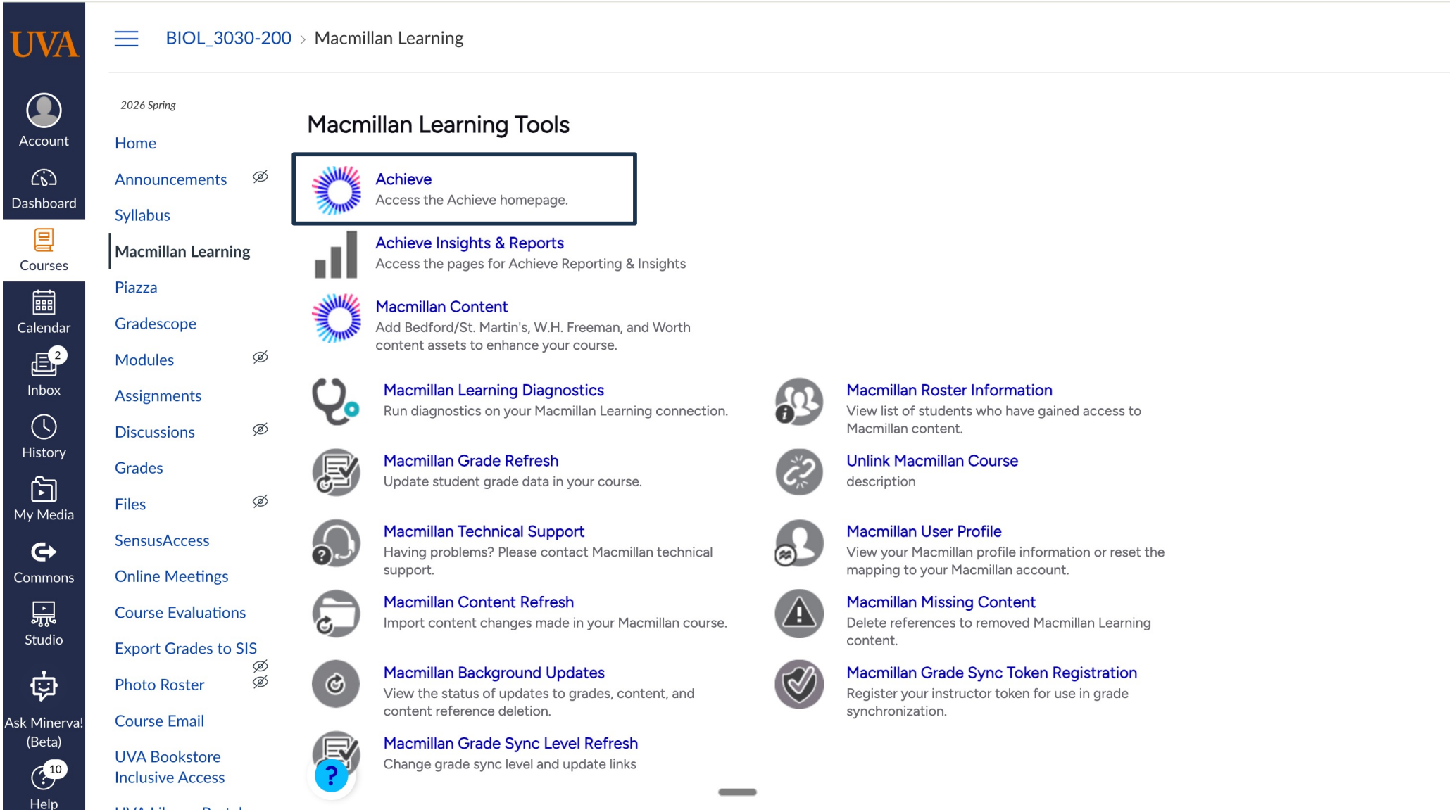
Music

https://open.spotify.com/artist/7AAFUvlYJtmuGBOcoqSMfh?si=TG8CAbiGROCpfE2ovVgv-Q

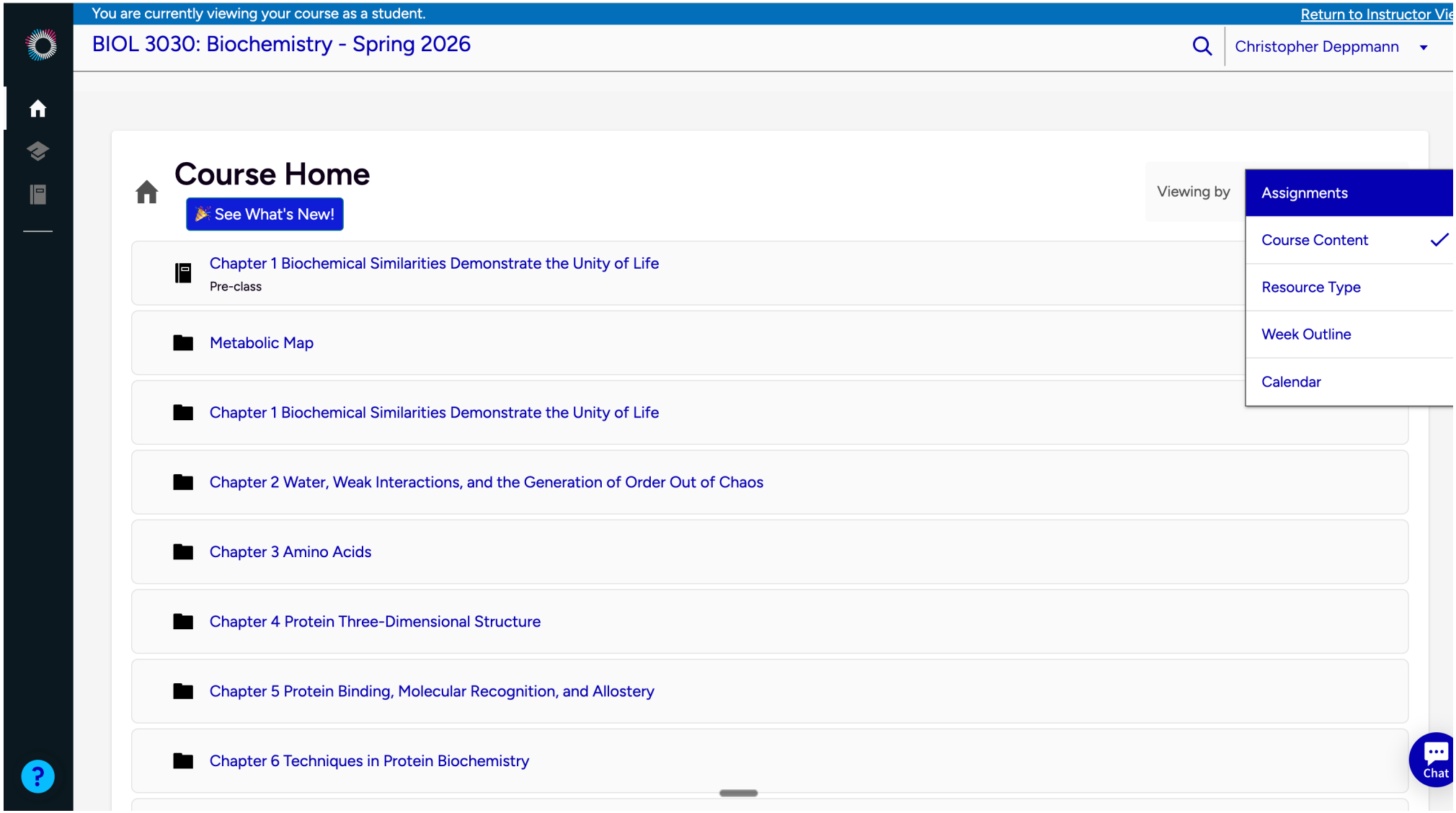
Rapid tour of our online environment.

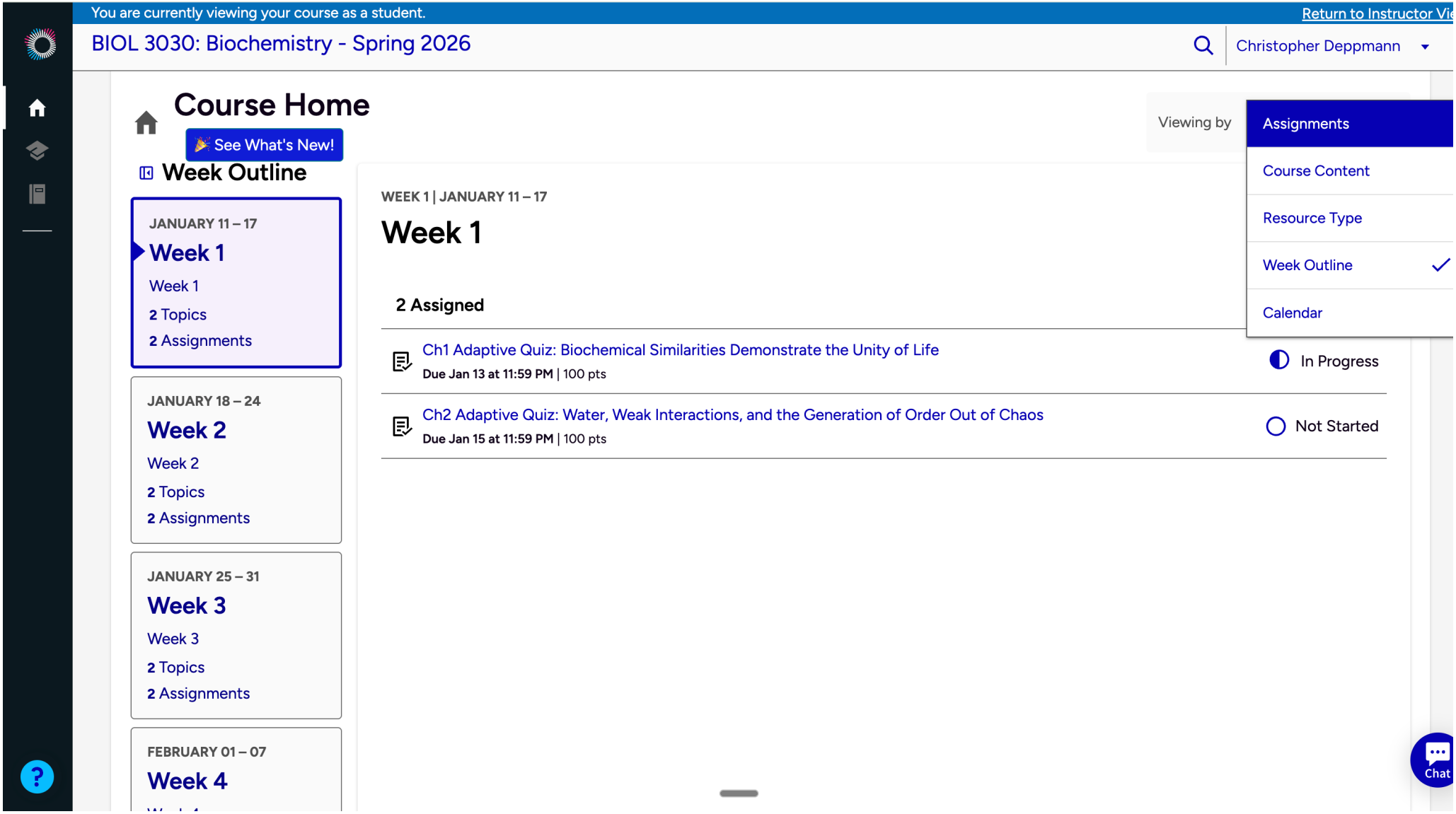
Getting Your textbook and online assignments



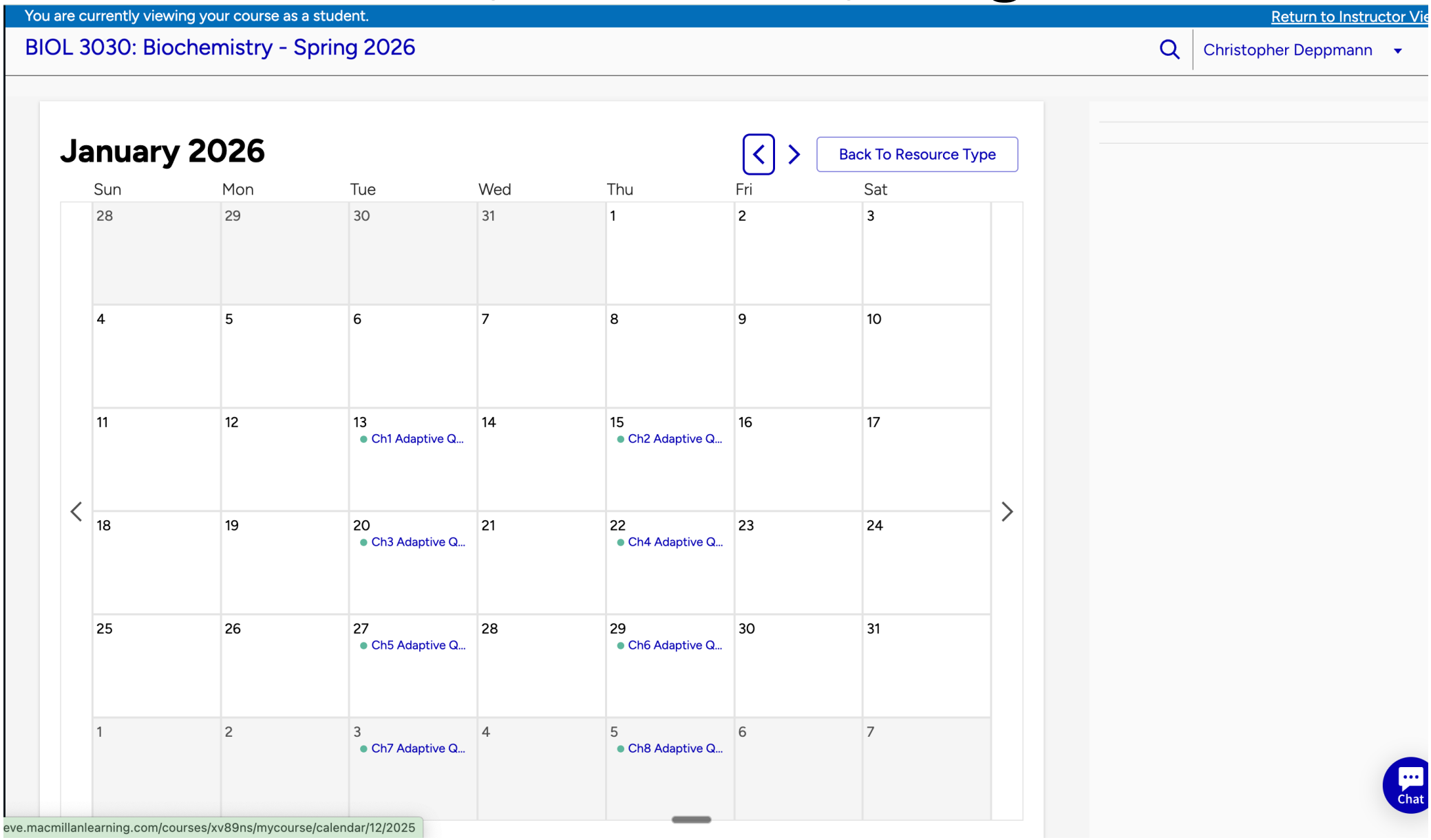

UVA
Account
Dashboard
Courses
Calendar
Inbox
History
My Media
Commons
Studio
Ask Minerva! (Beta)
Help
BIOL_3030-200 › Macmillan Learning
2026 Spring
Home
Announcements
Syllabus
Macmillan Learning
Piazza
Gradescope
Modules
Assignments
Discussions
Grades
Files
SensusAccess
Online Meetings
Course Evaluations
Export Grades to SIS
Photo Roster
Course Email
UVA Bookstore Inclusive Access
Macmillan Learning Tools
Achieve
Access the Achieve homepage.
Achieve Insights & Reports
Access the pages for Achieve Reporting & Insights
Macmillan Content
Add Bedford/St. Martin's, W.H. Freeman, and Worth content assets to enhance your course.
Macmillan Learning Diagnostics
Run diagnostics on your Macmillan Learning connection.
Macmillan Grade Refresh
Update student grade data in your course.
Macmillan Technical Support
Having problems? Please contact Macmillan technical support.
Macmillan Content Refresh
Import content changes made in your Macmillan course.
Macmillan Background Updates
View the status of updates to grades, content, and content reference deletion.
Macmillan Grade Sync Level Refresh
Change grade sync level and update links
Macmillan Roster Information
View list of students who have gained access to Macmillan content.
Unlink Macmillan Course
description
Macmillan User Profile
View your Macmillan profile information or reset the mapping to your Macmillan account.
Macmillan Missing Content
Delete references to removed Macmillan Learning content.
Macmillan Grade Sync Token Registration
Register your instructor token for use in grade synchronization.

Textbook, homework, assignments



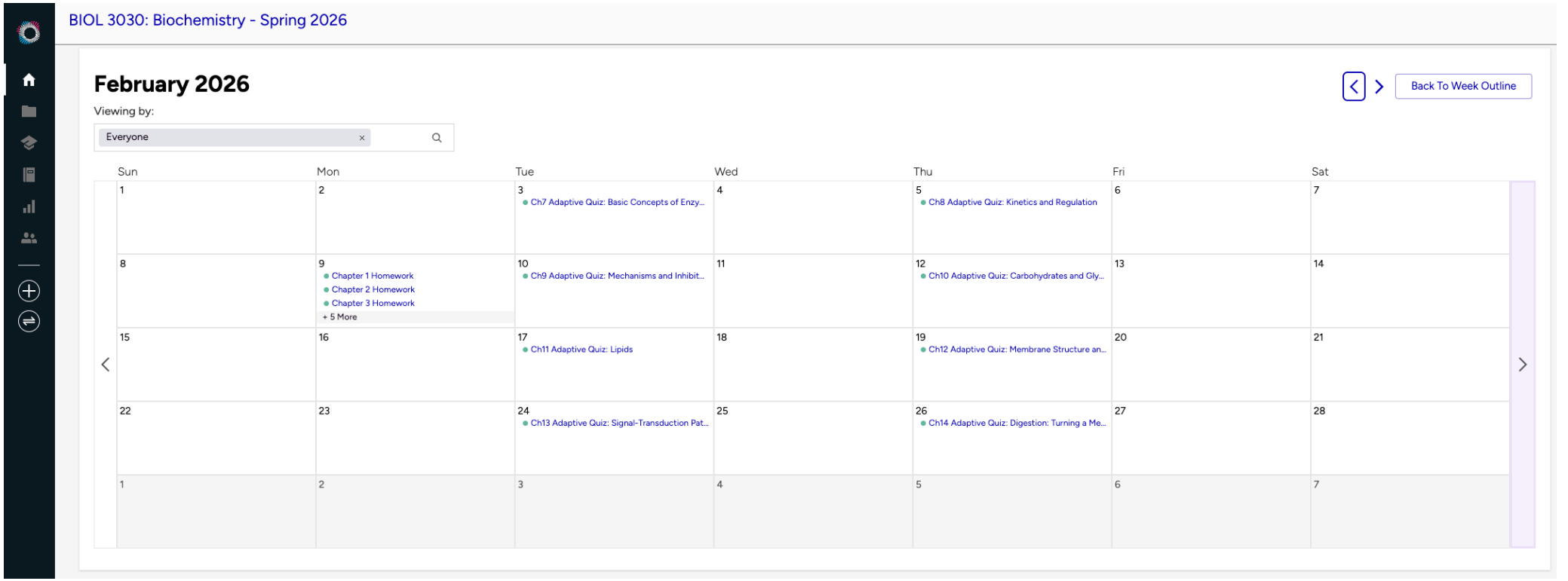

You are currently viewing your course as a student.
Return to Instructor Vie
BIOL 3030: Biochemistry - Spring 2026
Christopher Deppmann
Course Home
See What's New!
Week Outline
JANUARY 11 – 17
Week 1
Week 1
2 Topics
2 Assignments
JANUARY 18 – 24
Week 2
Week 2
2 Topics
2 Assignments
JANUARY 25 – 31
Week 3
Week 3
2 Topics
2 Assignments
FEBRUARY 01 – 07
Week 4
Viewing by
Assignments
Course Content
Resource Type
Week Outline
Calendar
WEEK 1 | JANUARY 11 – 17
Week 1
2 Assigned
Ch1 Adaptive Quiz: Biochemical Similarities Demonstrate the Unity of Life
Due Jan 13 at 11:59 PM | 100 pts
In Progress
Ch2 Adaptive Quiz: Water, Weak Interactions, and the Generation of Order Out of Chaos
Due Jan 15 at 11:59 PM | 100 pts
Not Started
Chat

Textbook, homework, assignments

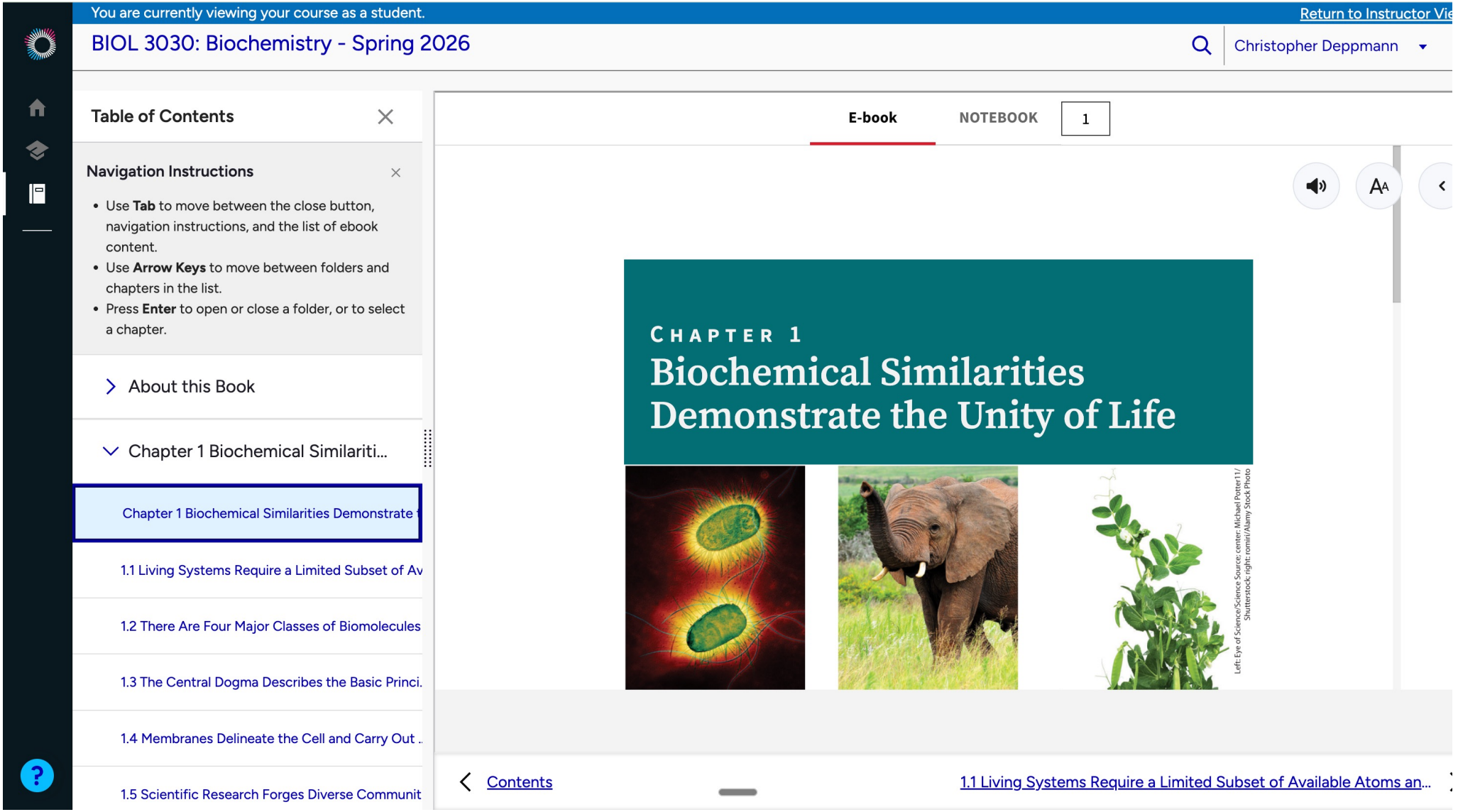


Note: your first quiz is due today.

Textbook, homework, assignments

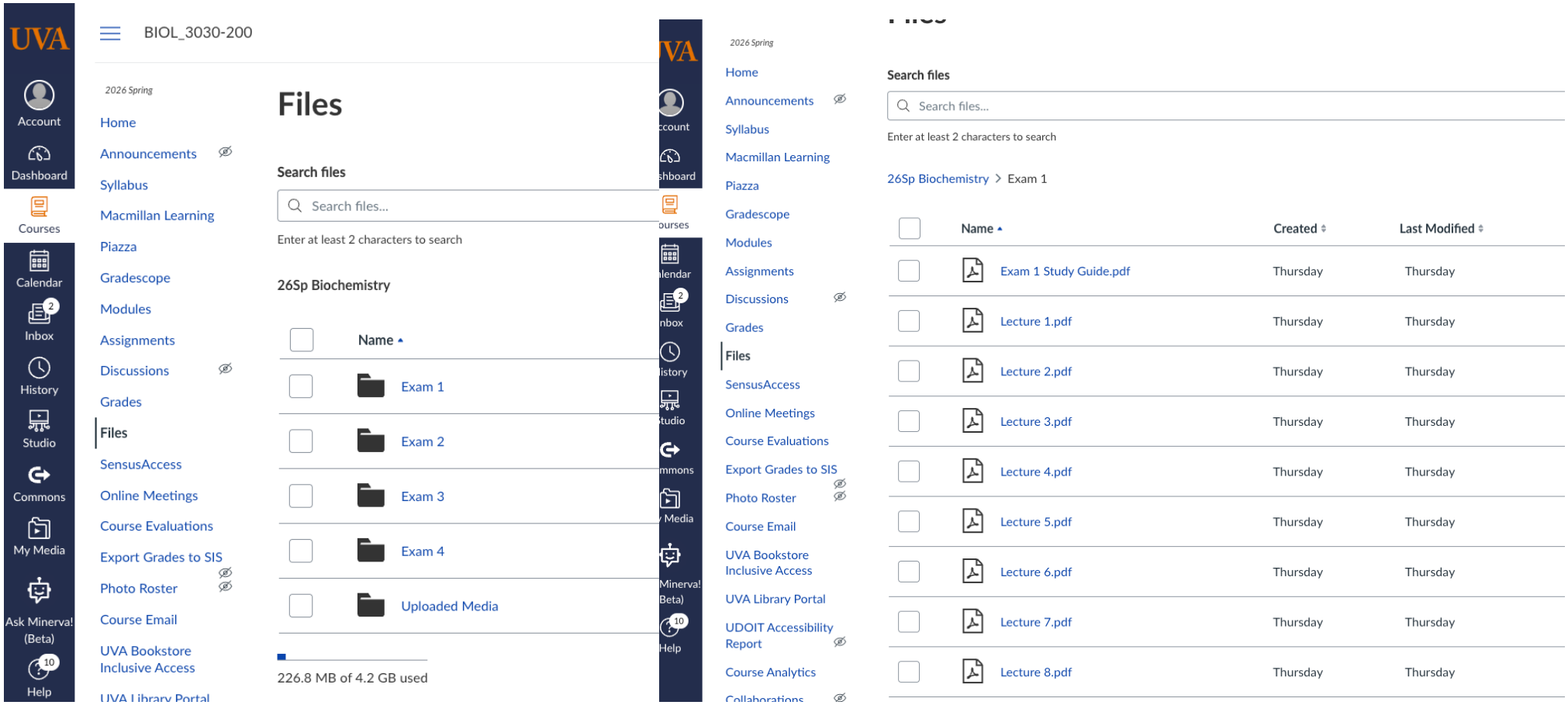


Textbook, homework, assignments



Resources on canvas-slides

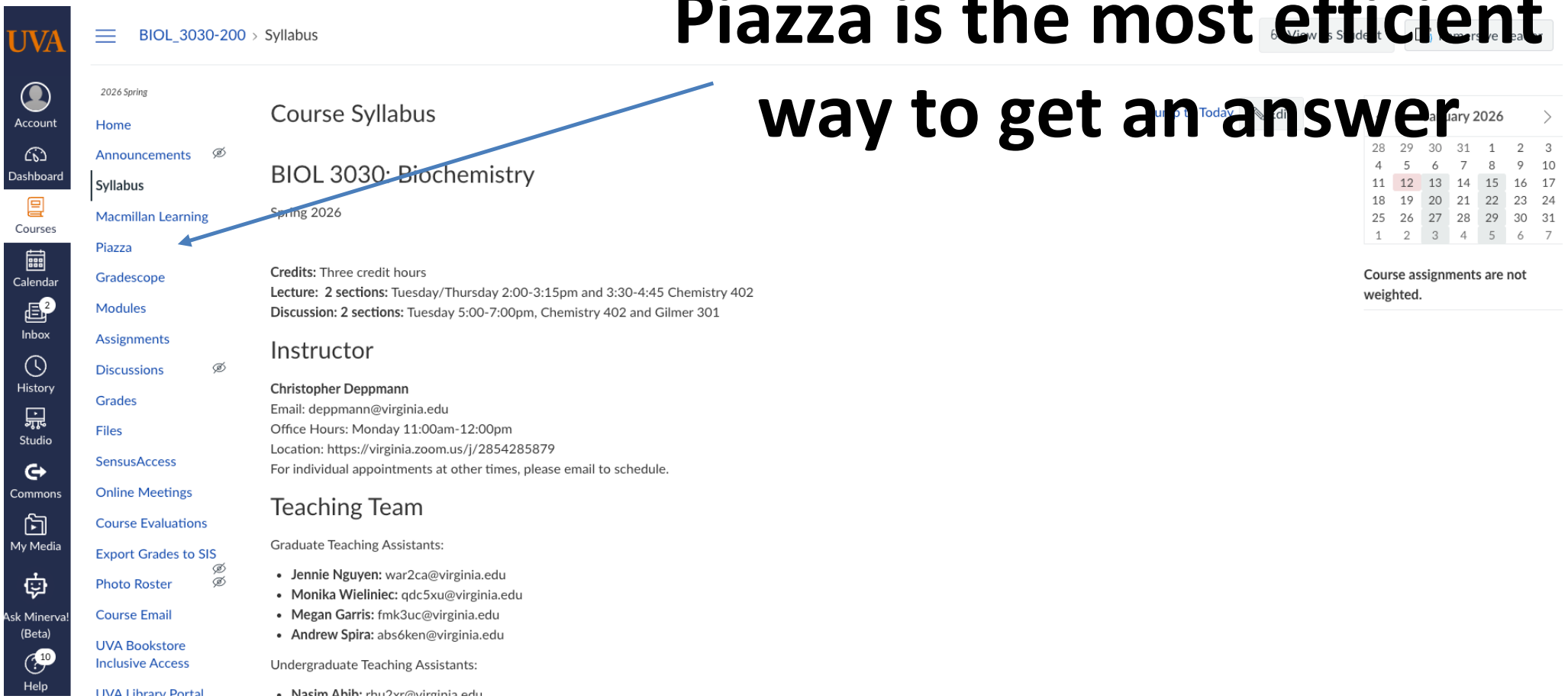
I'll post slides resources related to the chapters

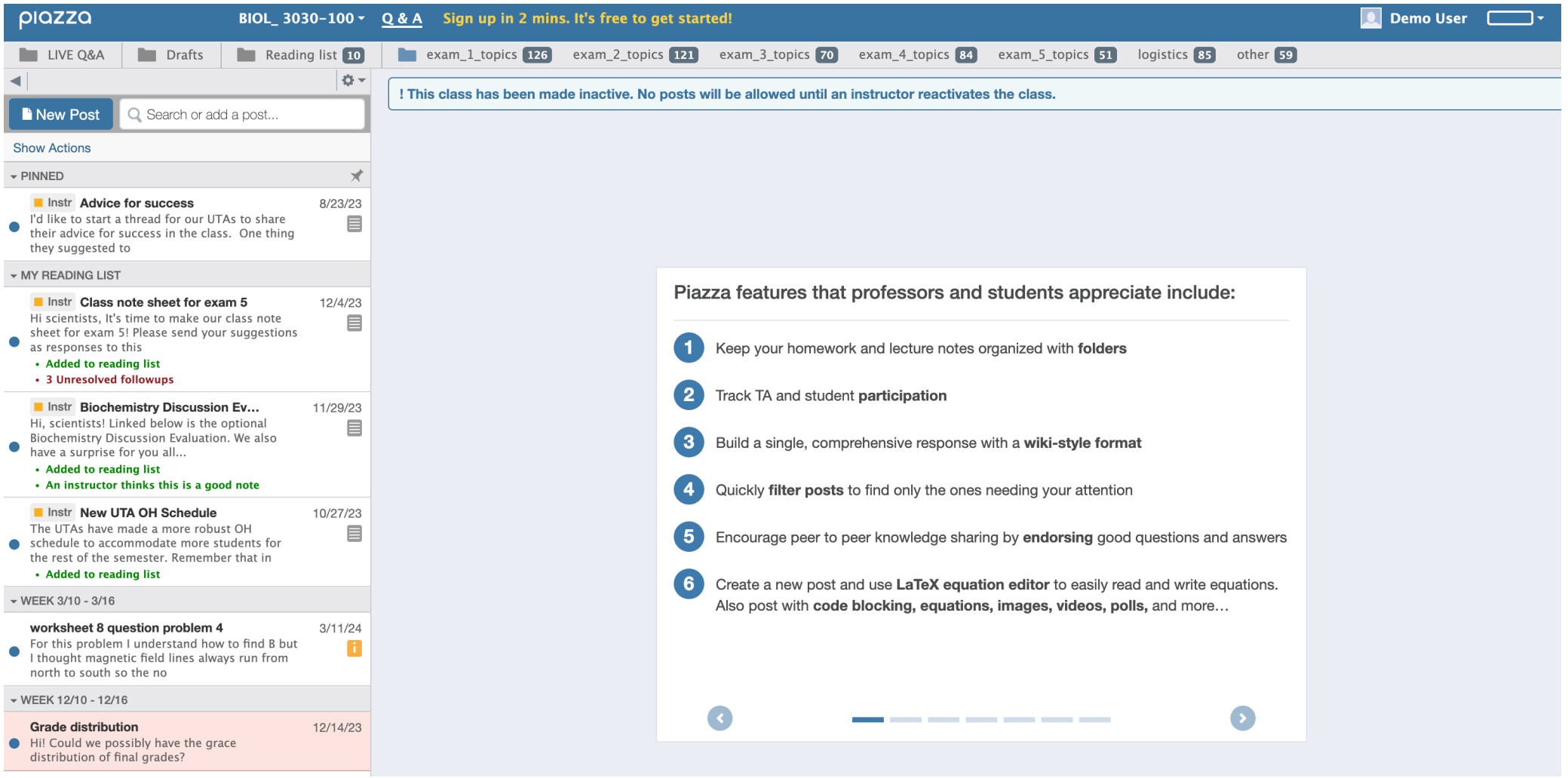


Note: Files are subject to change before a lecture.

Resources on canvas-piazza

Piazza is the most efficient way to get an answer



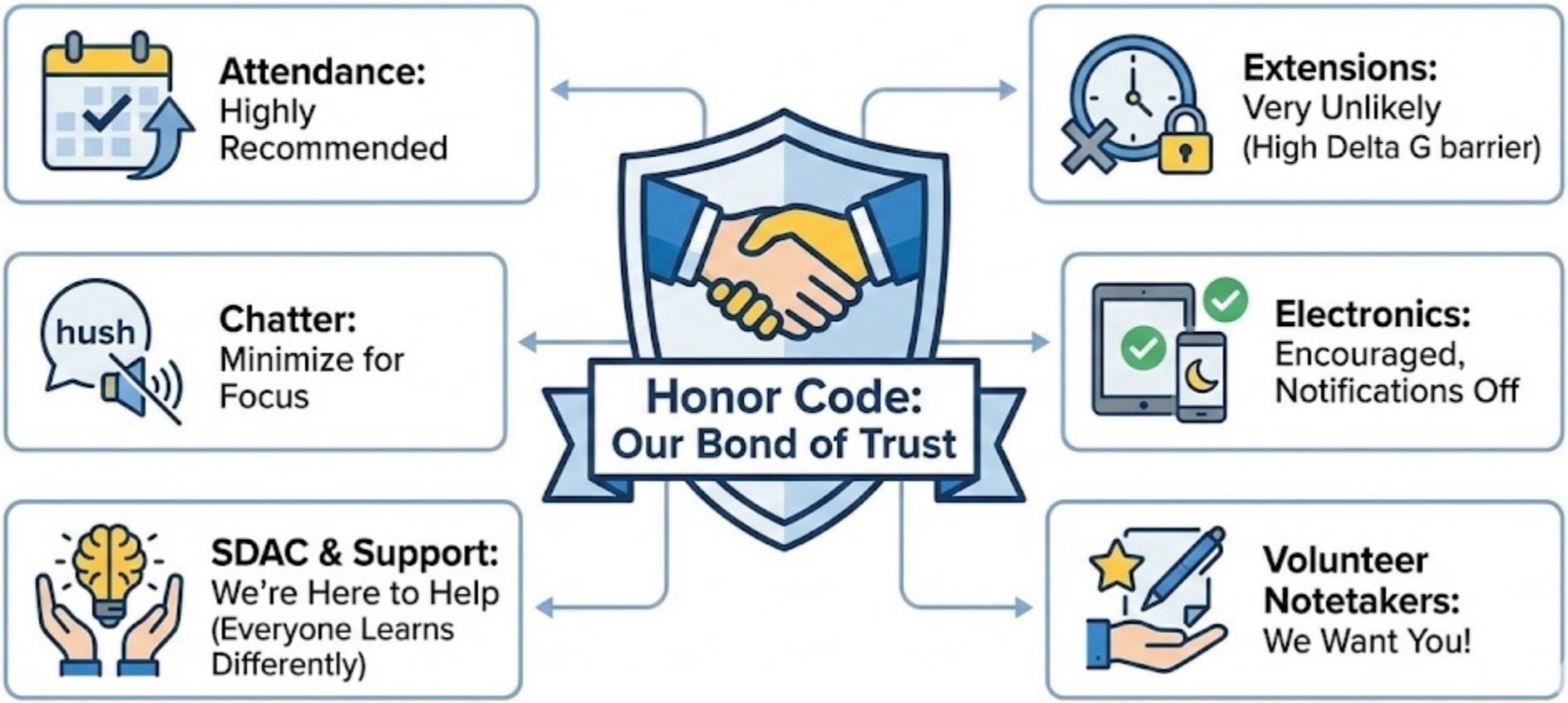


Piazza Tips:
Check for Existing Answers before posting.
Be Clear and Concise in your questions and responses.
Help Each Other by answering questions—it's a great way to learn!
Use Descriptive Titles for easy navigation.
Stay Engaged with notifications and follow-ups.

WEEK 1 CHECKLIST & LAUNCH

- ◯ Access Achieve via Canvas
- ◯ Read Chapter 1
- ◯ Complete Reading Quiz
- ◯ Join Piazza
- ◯ Add Exam Dates to Calendar
- ◯ Decide on Inclusive Access
- ◯ ❗ Note: Discussions Start Week 2

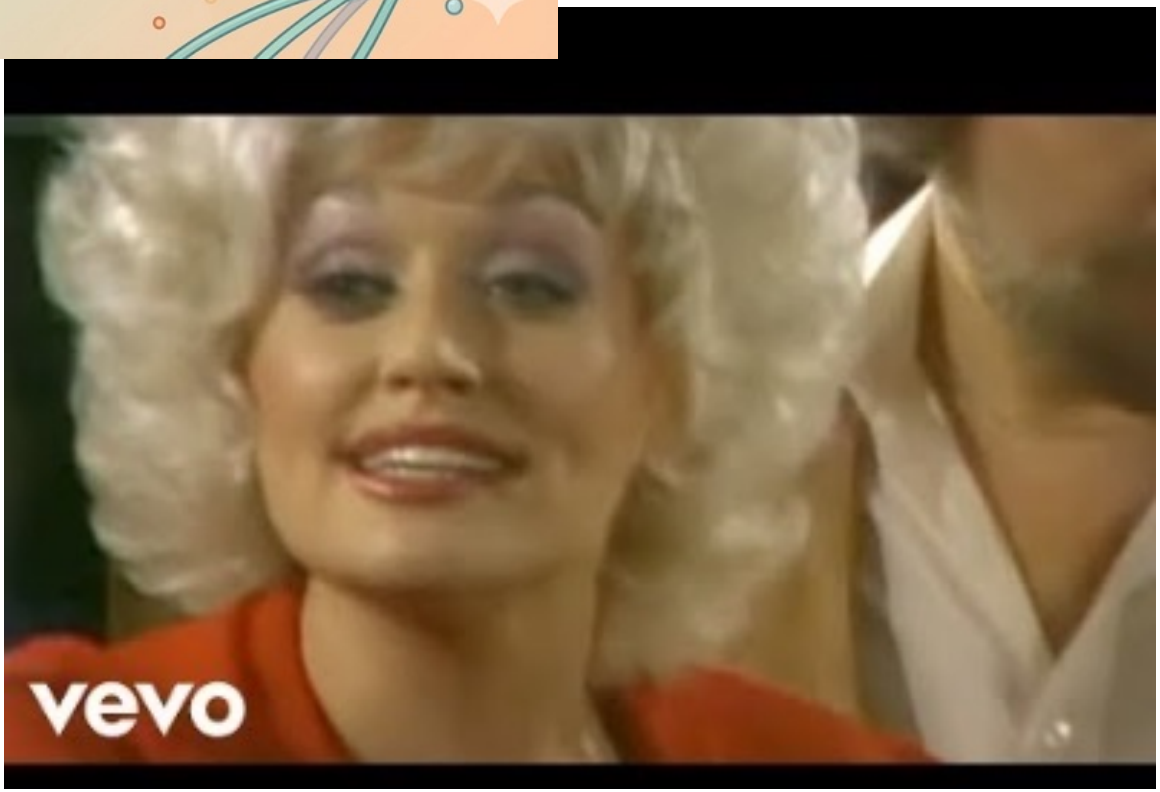
Course Logistics & Policies



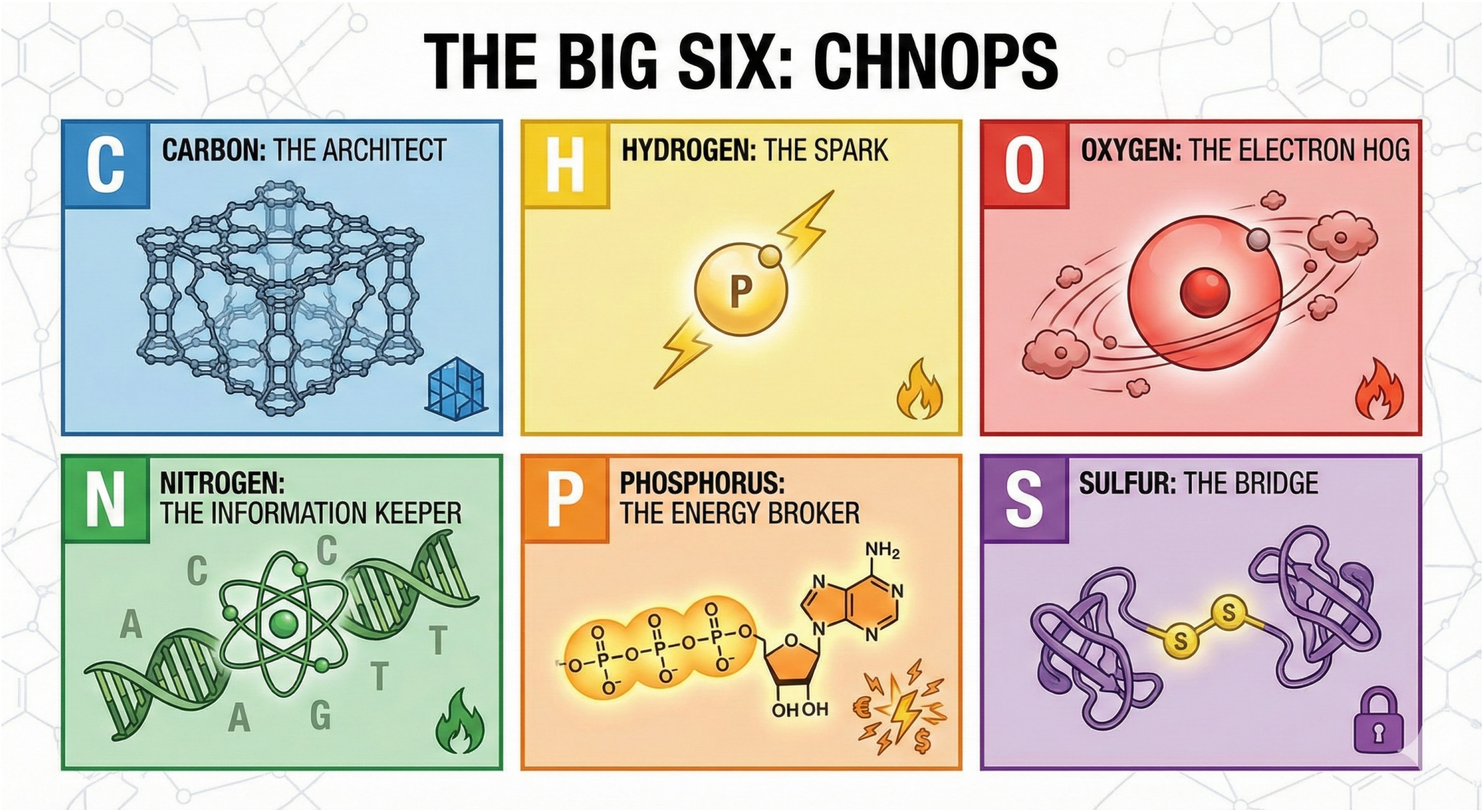
Academic Integrity

Questions?

BREAK - Back in 5 minutes
5:00
Even ATP needs to recharge!


vevo

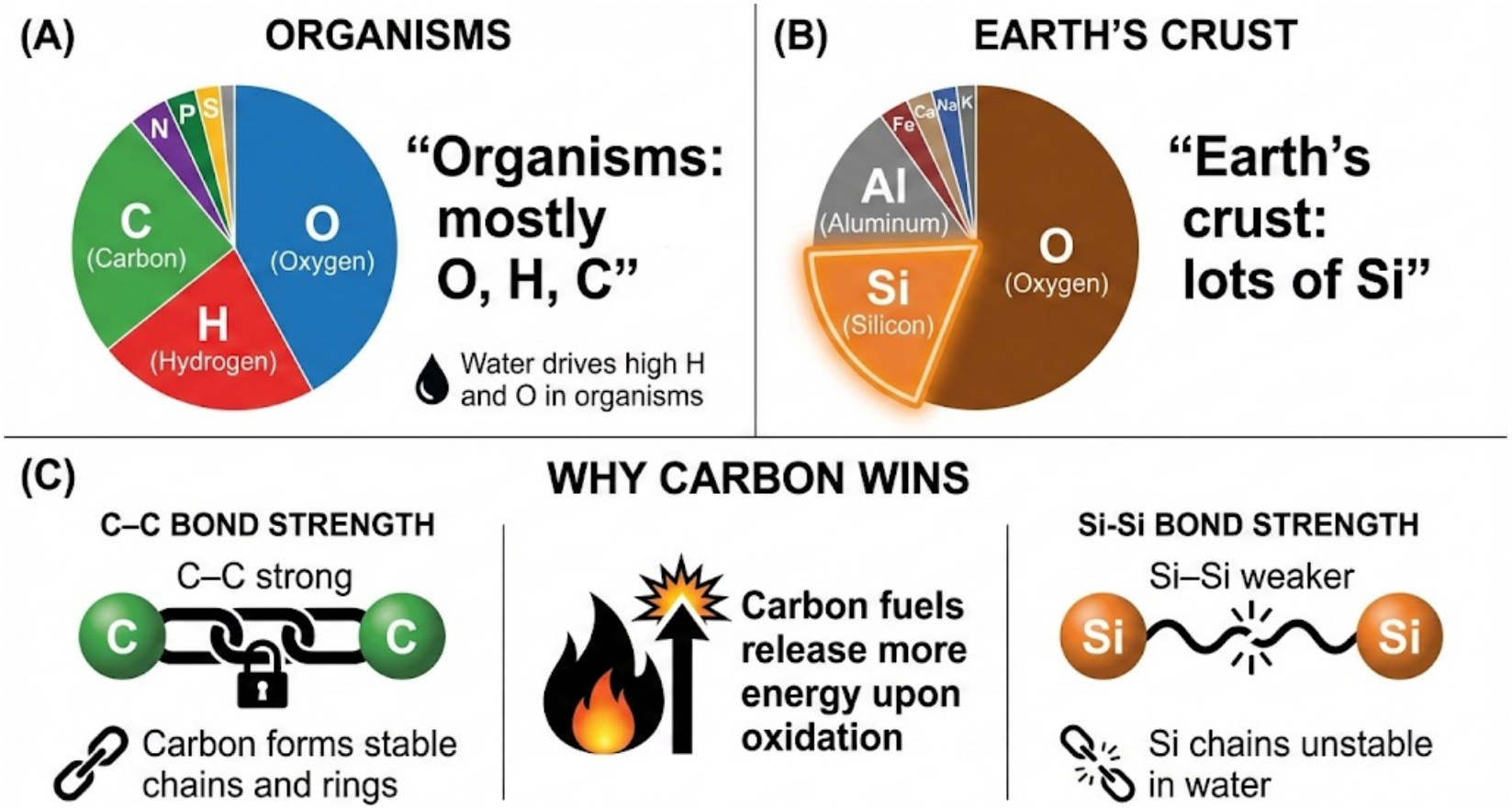
Chapter 1

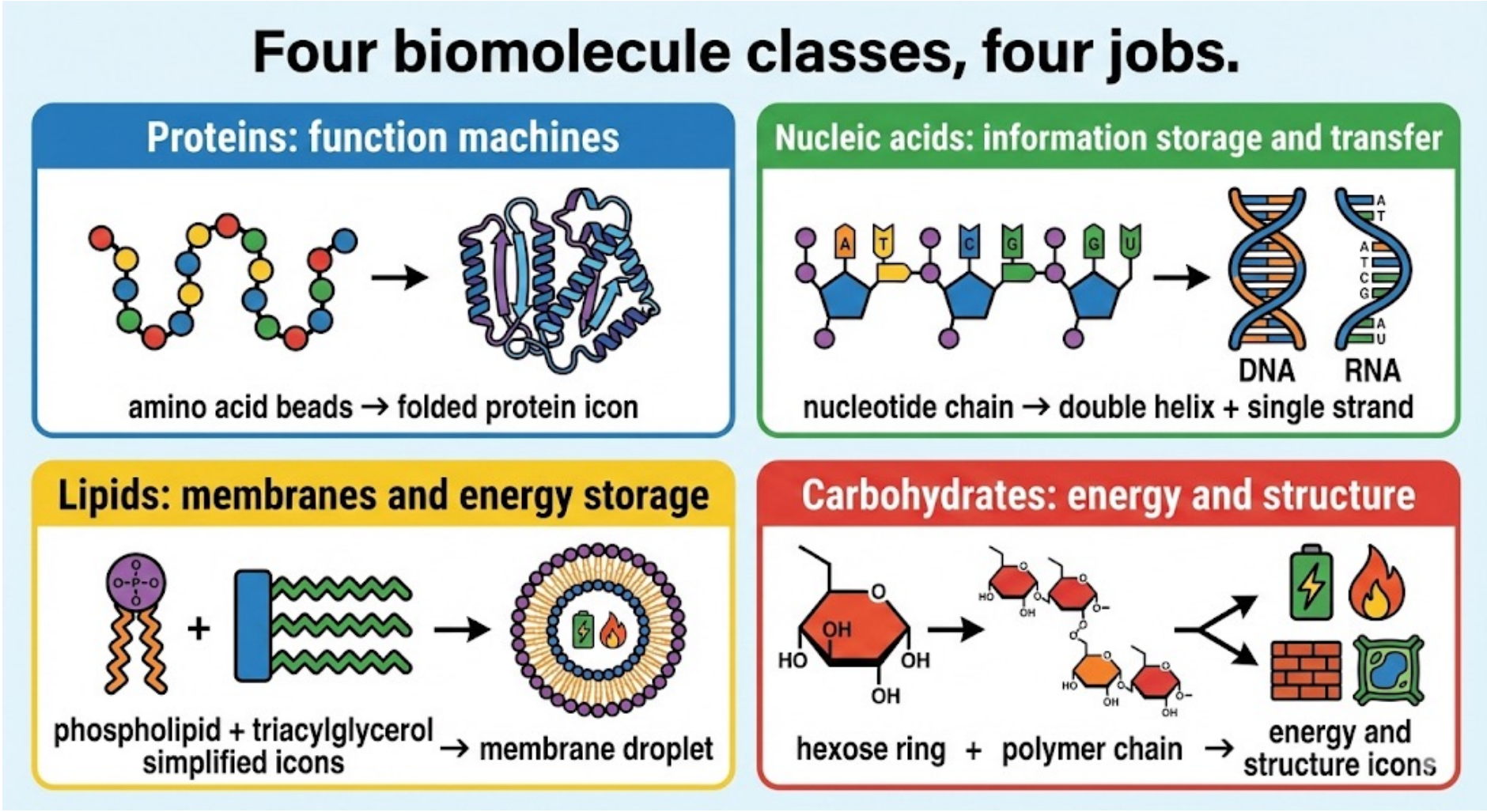

THE BIG SIX: CHNOPS
C
CARBON: THE ARCHITECT
H
HYDROGEN: THE SPARK
P
O
OXYGEN: THE ELECTRON HOG
N
NITROGEN:
THE INFORMATION KEEPER
C
C
A
T
T
A
G
P
PHOSPHORUS:
THE ENERGY BROKER
NH2
OH OH
S
SULFUR: THE BRIDGE
S
S

Why carbon?

Section 1.1 Living Systems Require a Limited Variety of Atoms and Molecules

- Of the 90 naturally occurring elements, only three—oxygen, hydrogen, and carbon—make up 98% of the atoms in any organism.
- Hydrogen and oxygen are so prevalent because of the ubiquity of water.
- Carbon is uniquely suited to be a key atom of biomolecules.





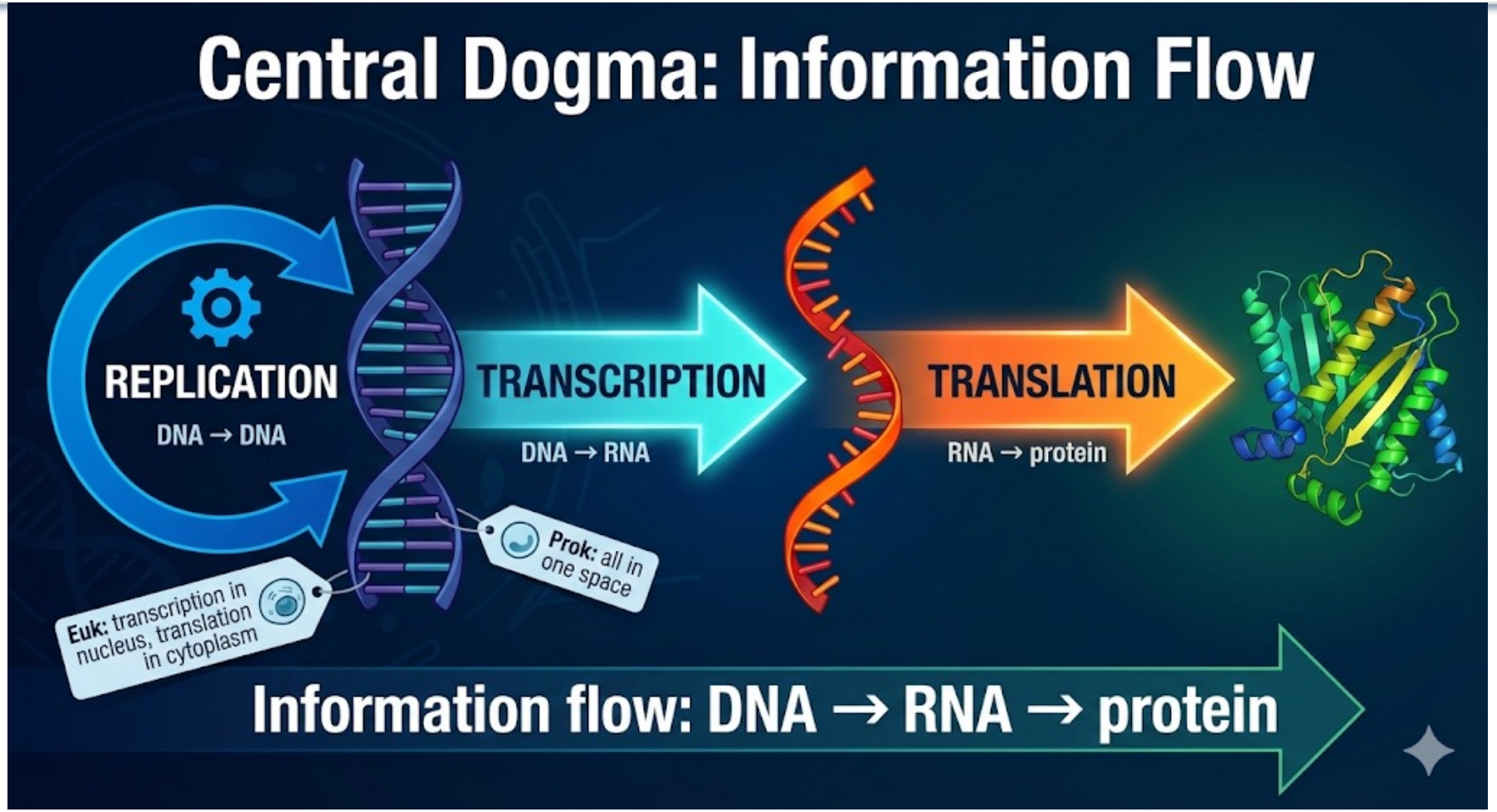
Section 1.2 There are Four Major Classes of Biomolecules

Learning Objective 1: Describe the key classes of biomolecules and differentiate between them.

Section 1.3 The Central Dogma Describes the Basic Principles of Biological Information Transfer

- **Learning Objective 2: List the steps of the central dogma.**

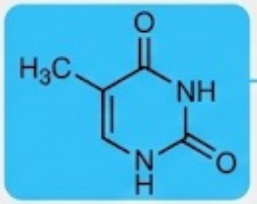
- The central dogma states that information flows from DNA to RNA to protein. Moreover, DNA is replicated.



DNA vs RNA Comparison

DNA

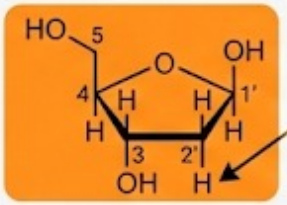
BASE



DNA uses **thymine (T)**

T ⋮⋮ A Thymine (T)

SUGAR



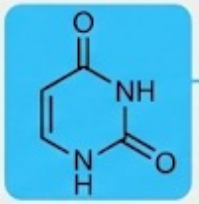
Deoxyribose (2' H)

Deoxyribose **(2' H)**

TYPICAL STRUCTURE

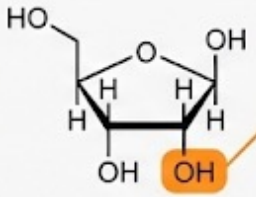
Double helix

RNA



RNA uses **uracil (U)**

T ⋮⋮ A Uracil (U)



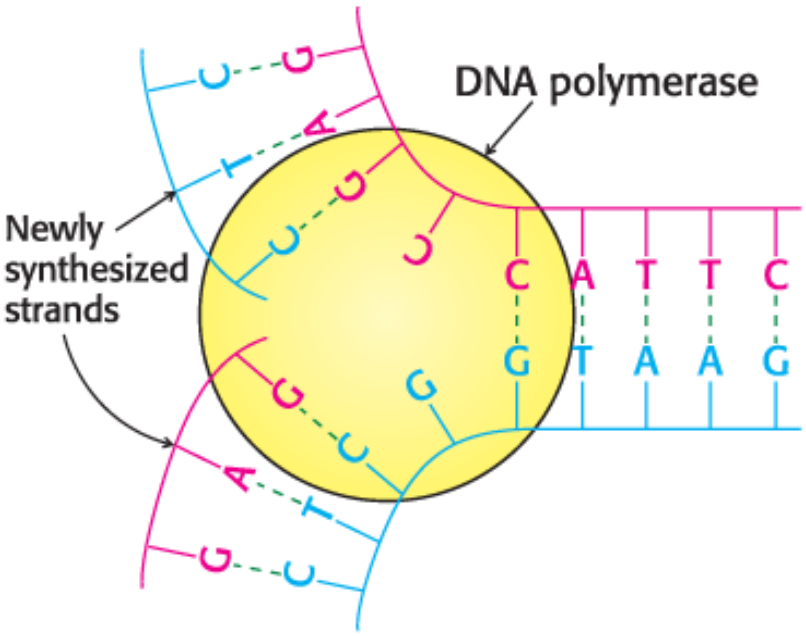
Ribose

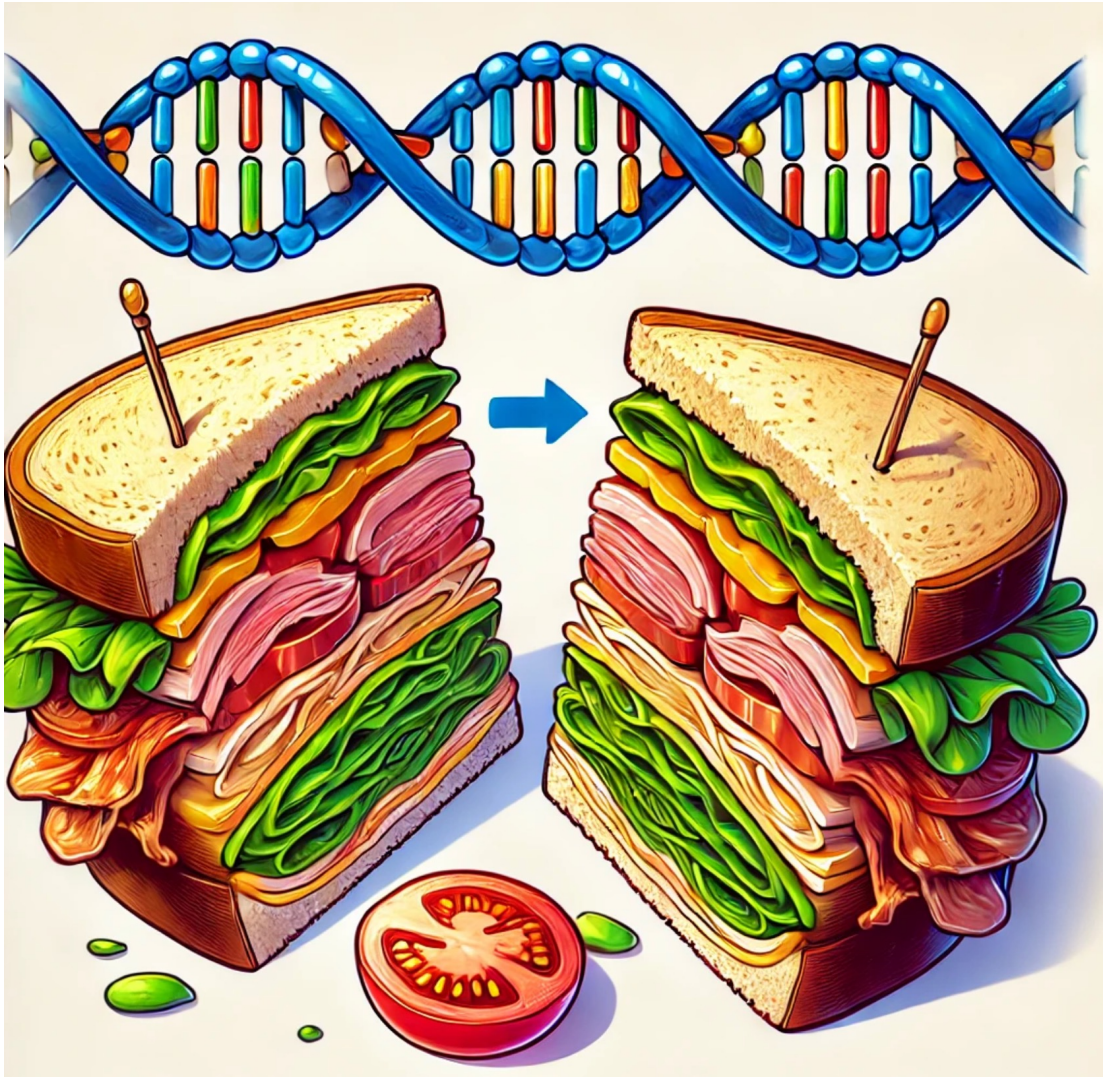
RNA has a **2' hydroxyl (2' OH)**

Single strand

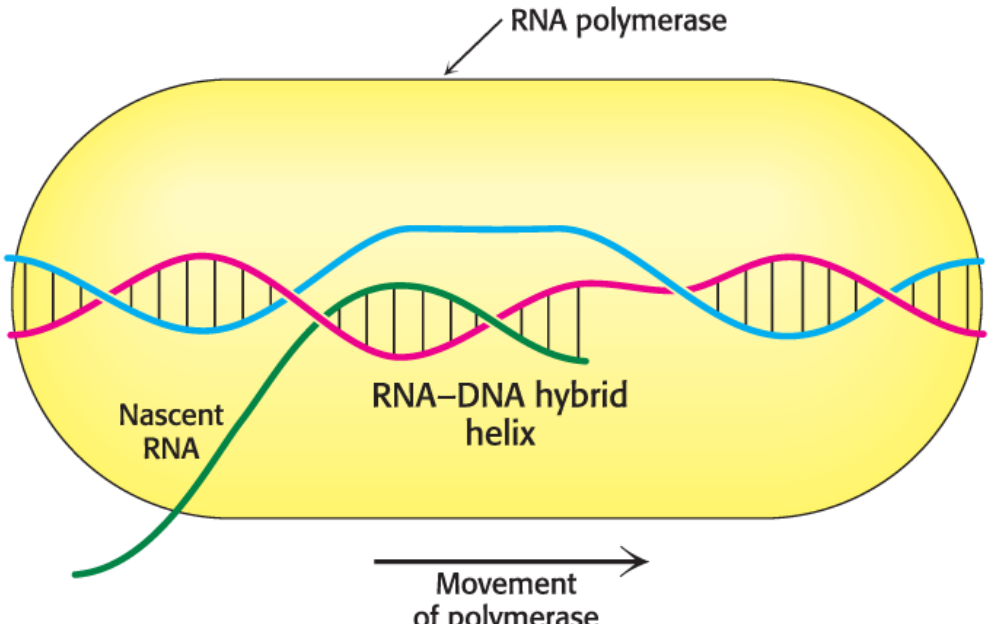
DNA Replication

- DNA is heritable information: the genome.
- DNA is replicated by a group of enzymes collectively called DNA polymerase.





Transcription



- **DID YOU KNOW?**
  - As defined in the Oxford English Dictionary, *to transcribe* means “to make a copy of (something) in writing; to copy out from an original; to write (a copy).”
- RNA polymerase catalyzes transcription: the process of copying DNA information into RNA.

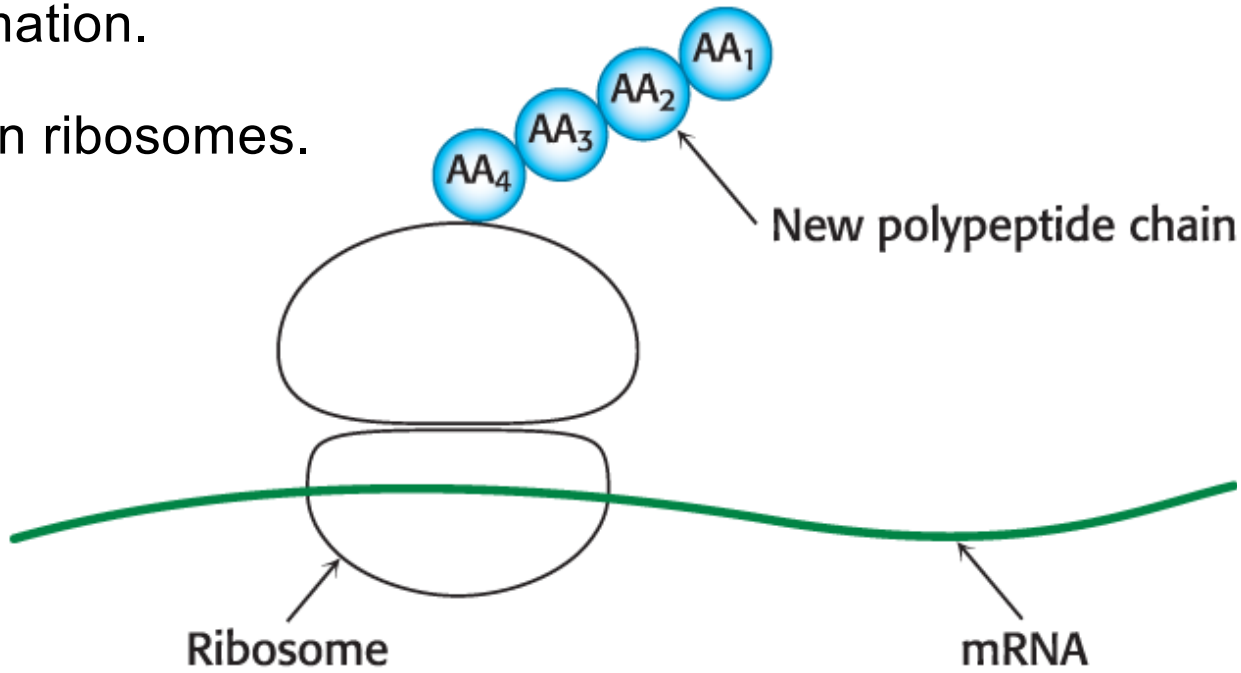
- Selective transcription of the genome defines the function of a cell or tissue.

mRNA (the messenger) tRNA (the transporter)

rRNA (the builder)

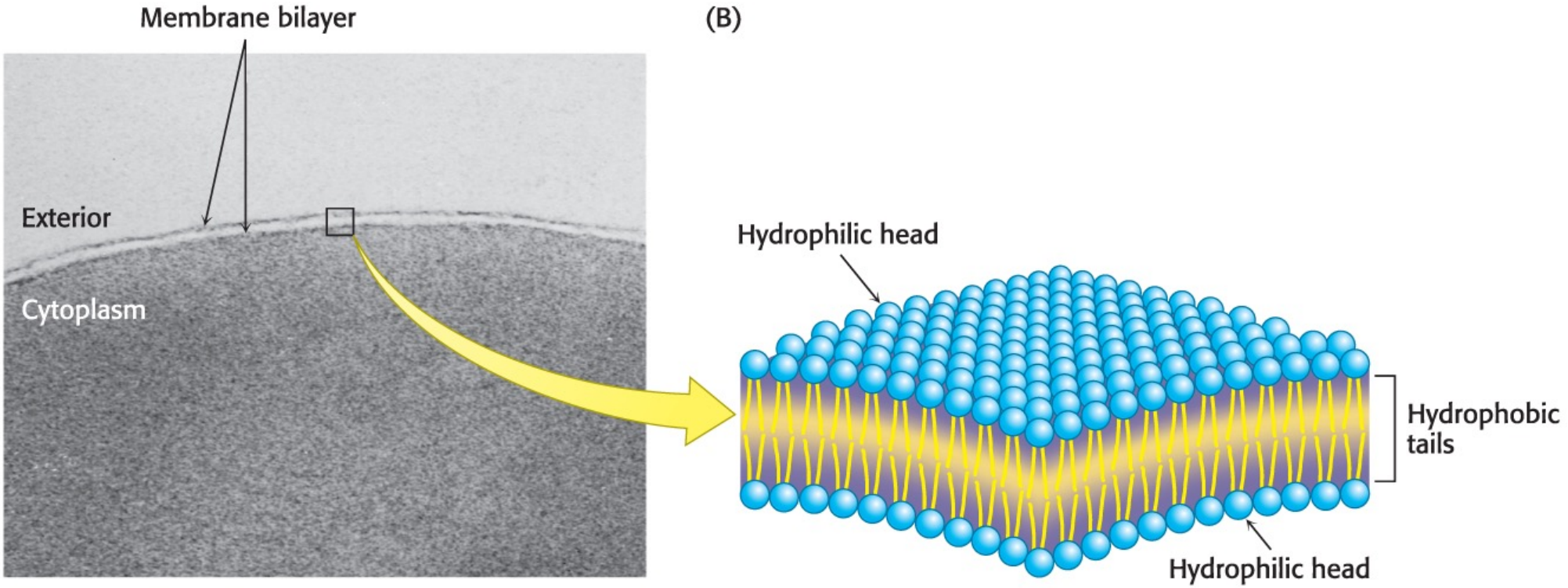
Translation

- Translation converts the nucleic acid sequence information in mRNA into protein sequence information.
- Translation occurs within ribosomes.



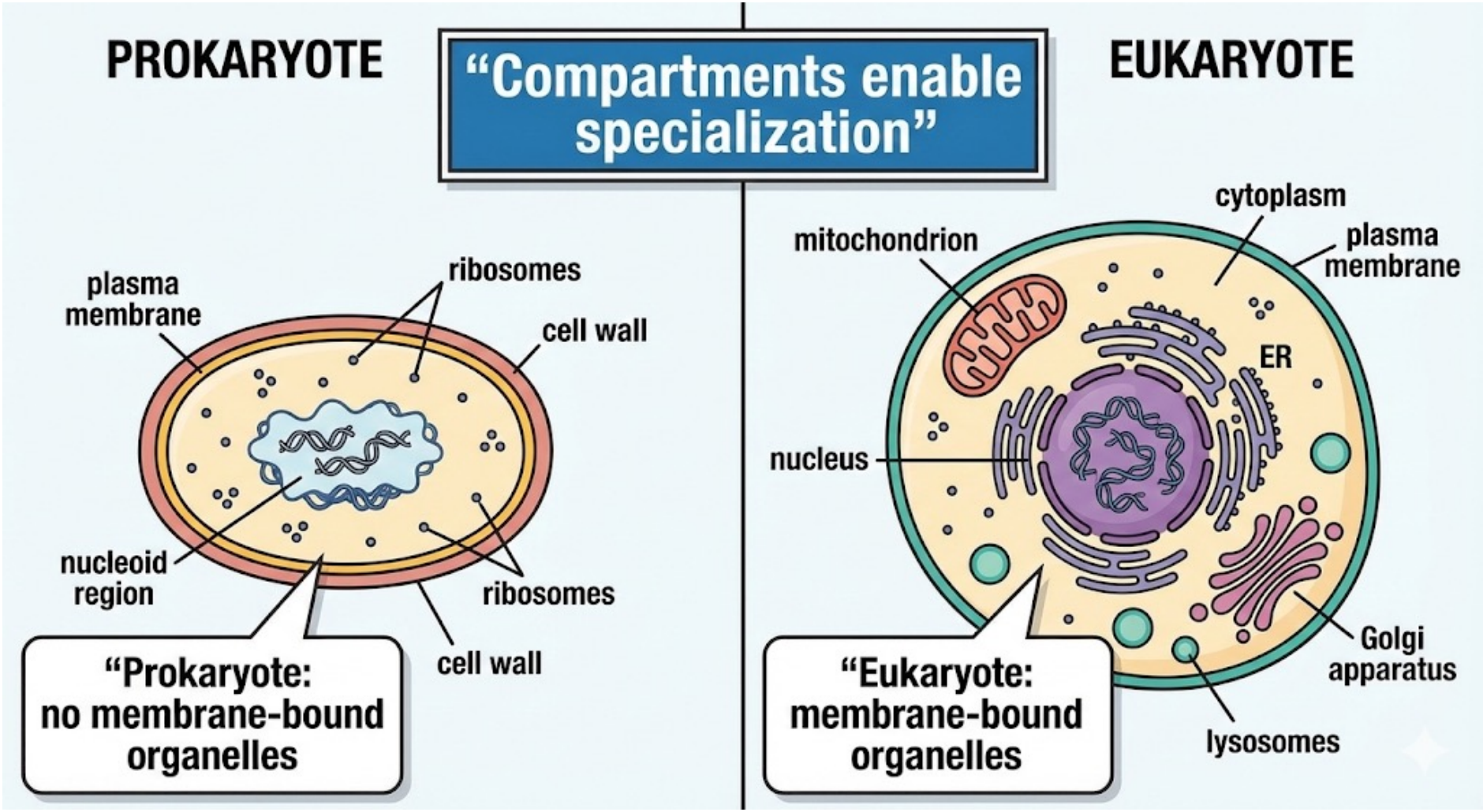
Section 1.4 Membranes Define the Cell and Carry Out Cellular Functions

- **Learning Objective 3: Identify the key features that differentiate eukaryotic cells from prokaryotic cells.**
- Eukaryotes contain membrane-enclosed compartments inside the cell.
- Prokaryotes lack intracellular membranes.



Quick check: are “your” cells all eukaryotic?

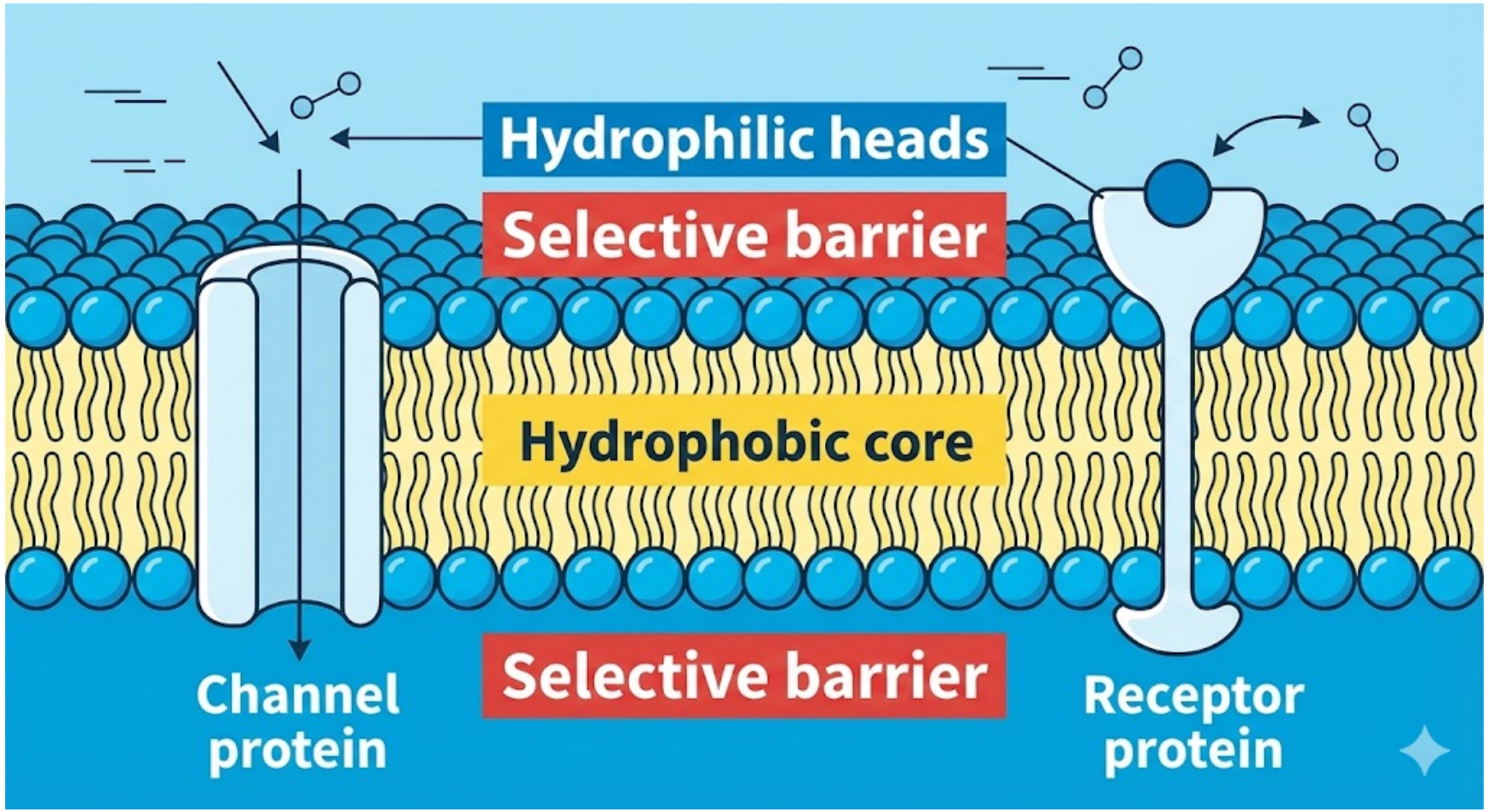
“The human body contains only eukaryotic cells.” True or False?



Prokaryotic and Eukaryotic Cells

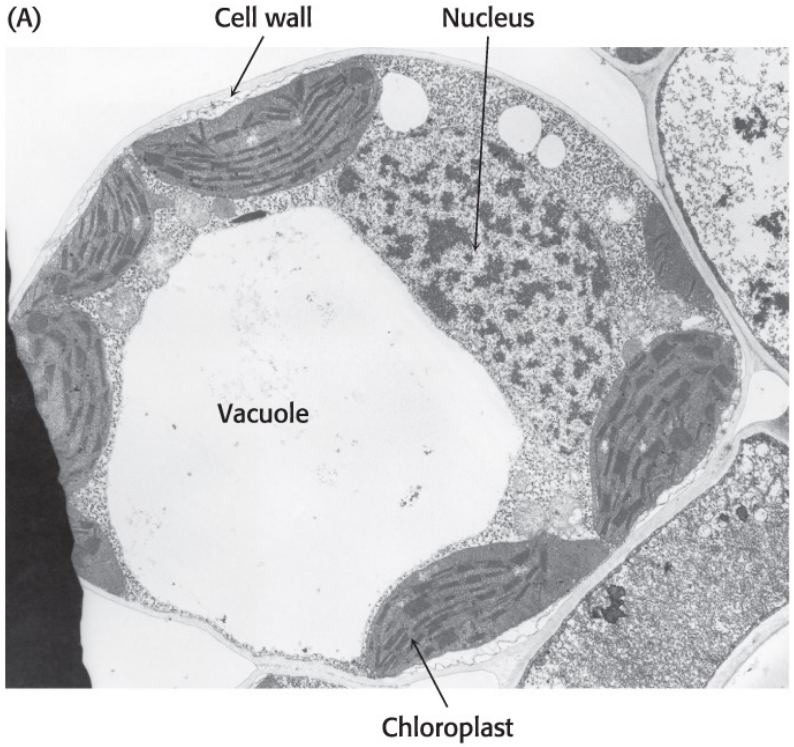
Plasma Membrane

- The plasma membrane separates the inside of the cell from the outside.
- The plasma membrane is impermeable to most biomolecules.
- Selective permeability occurs because of the presence of proteins associated with the membrane.

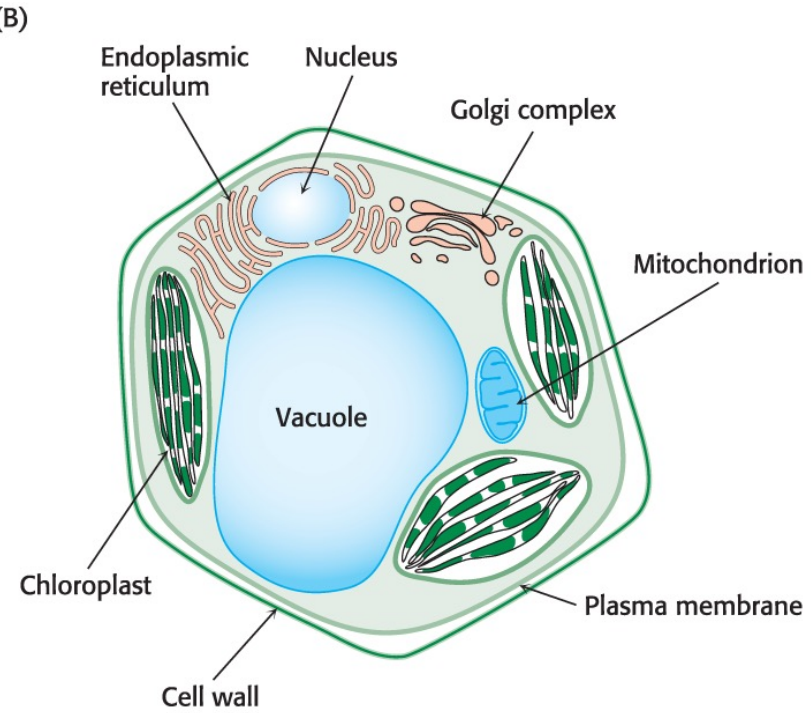


Plasma Membrane of a Plant Cell

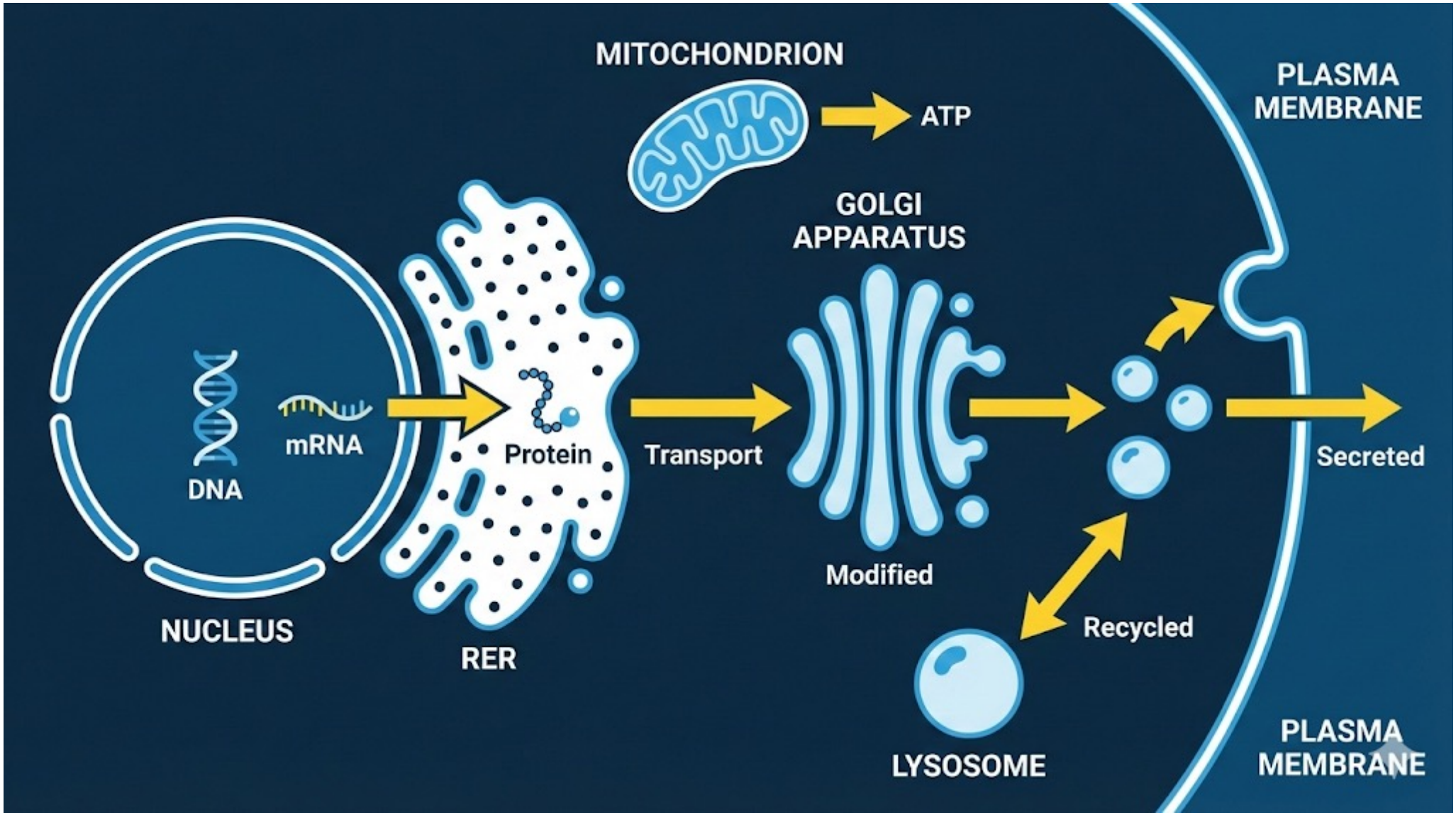
- The plasma membrane of a plant cell is surrounded by a cell wall composed largely of cellulose, a linear polymer of glucose.
- Large vacuoles, unique to plants, store water, ions, and various nutrients.



(A) Biophoto Associates/Science Source.

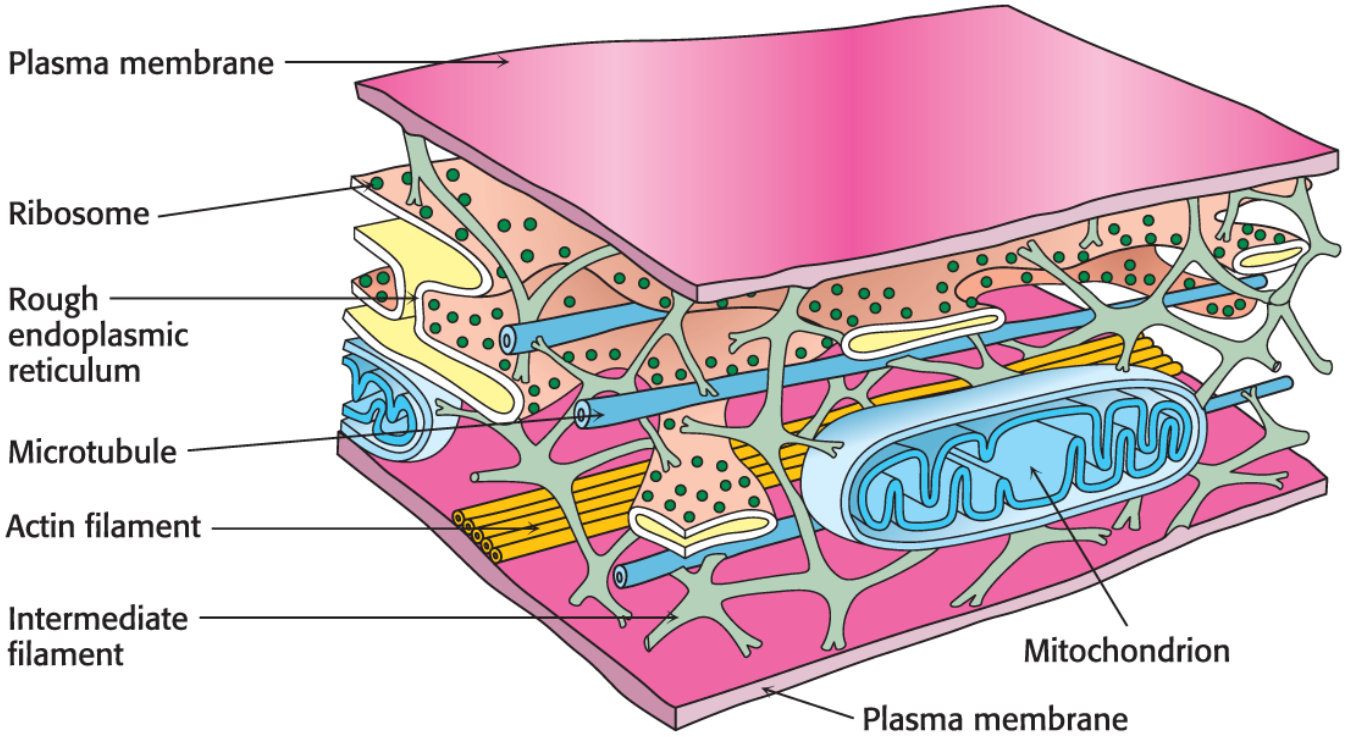


Organelles


MITOCHONDRION
ATP
PLASMA MEMBRANE
GOLGI APPARATUS
DNA
mRNA
Protein
Transport
Secreted
Modified
Recycled
NUCLEUS
RER
LYSOSOME
PLASMA MEMBRANE

Cytoplasm

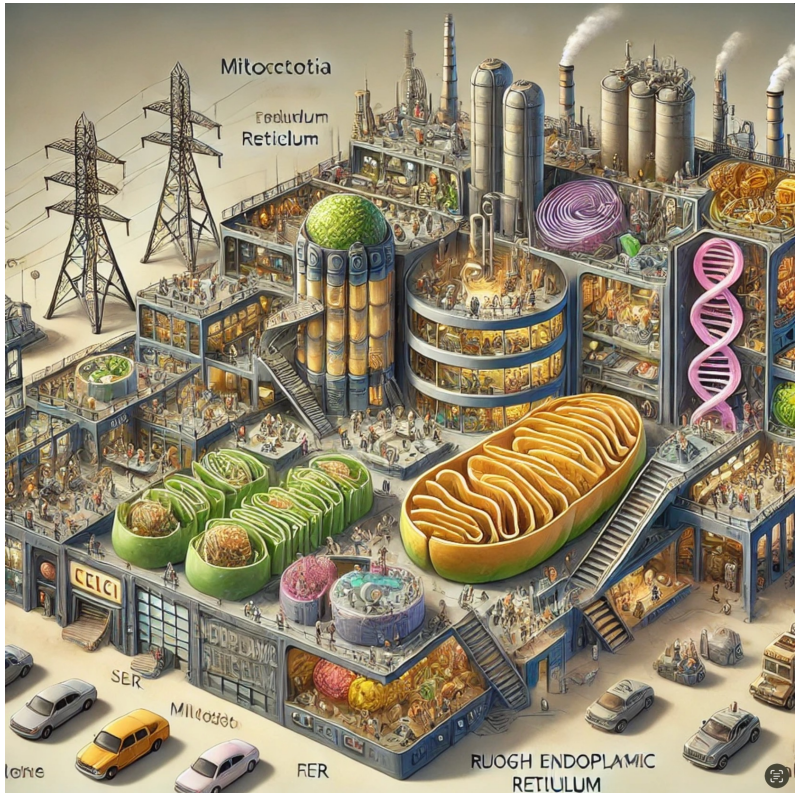
- The cytoplasm is the part of the cell surrounded by the plasma membrane but not enclosed by any intracellular membranes.
- The cytoplasm is organized by a series of structural filaments called the cytoskeleton.

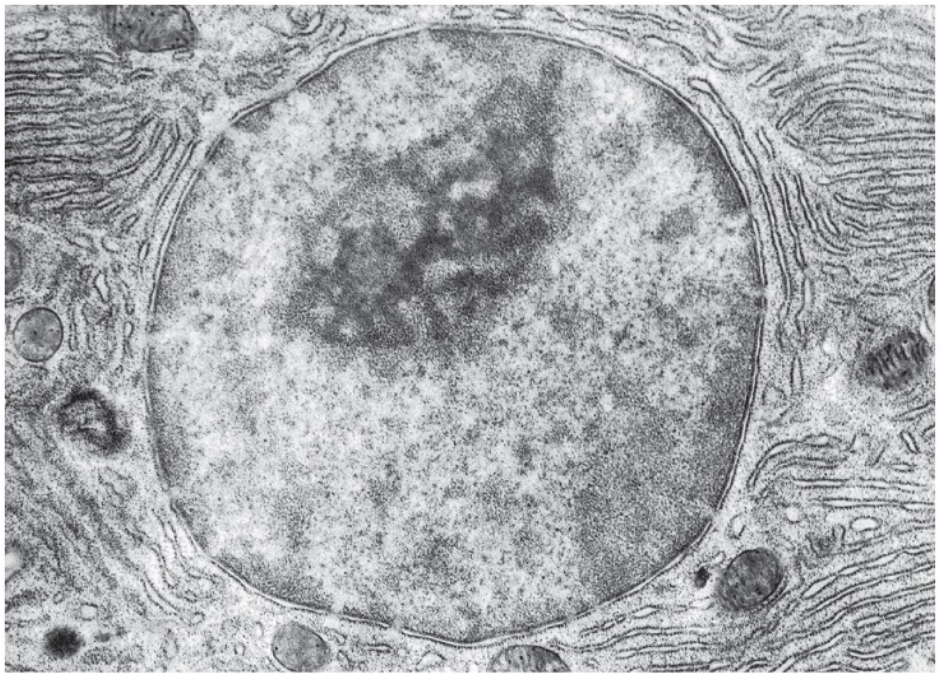


Tymoczko et al., *Biochemistry: A Short Course*, 4e, © 2019 W. H. Freeman and Company

Cellular Compartments

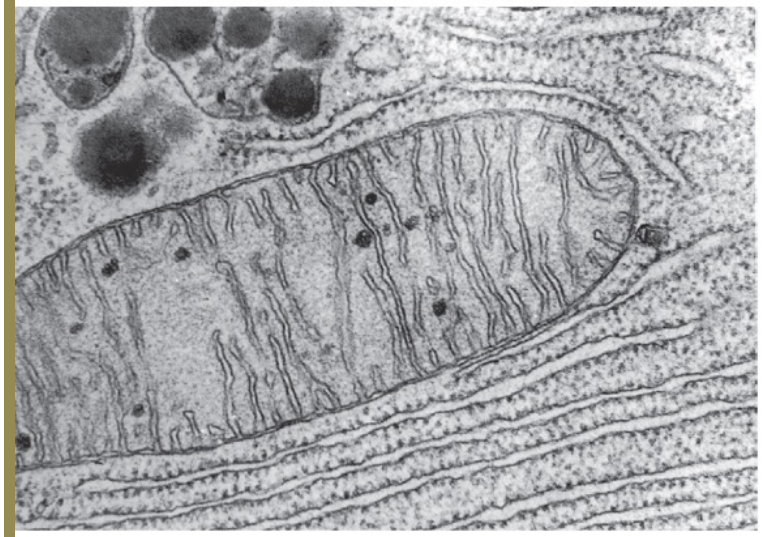
- Biochemical functions are sequestered in cellular compartments.
- The nucleus is the information center of the cell.



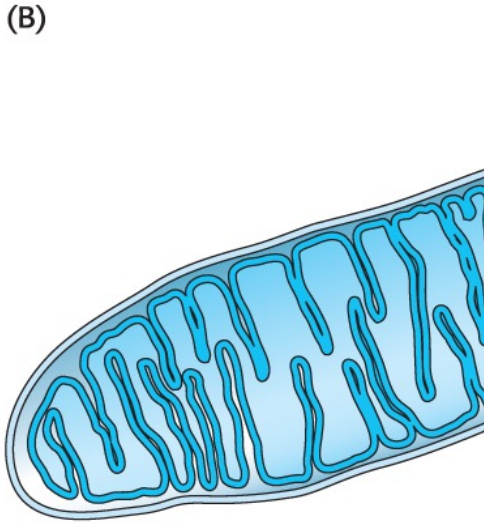


Don W. Fawcett/Science Source.

Mitochondria

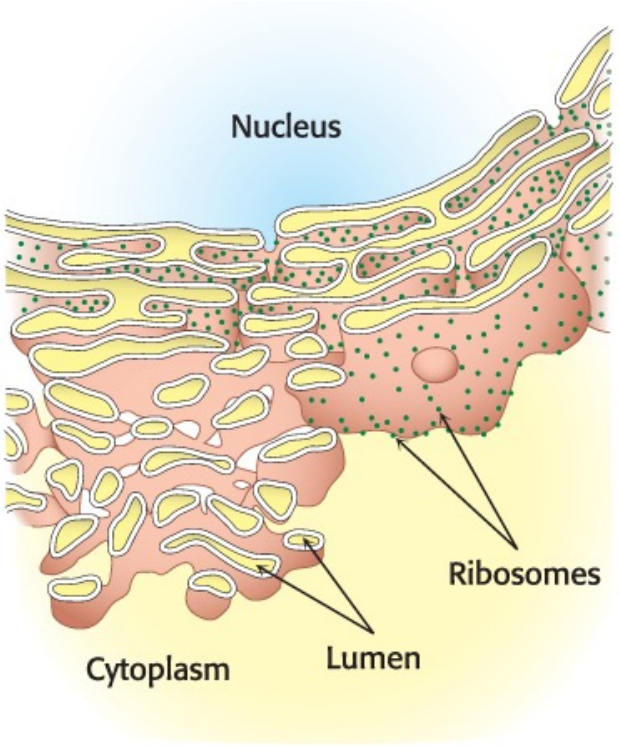


R. Porter/Science Source.



- Mitochondria are the primary site of ATP generation in eukaryotic cells.
- Chloroplasts, found in plant cells, are the site of photosynthesis.

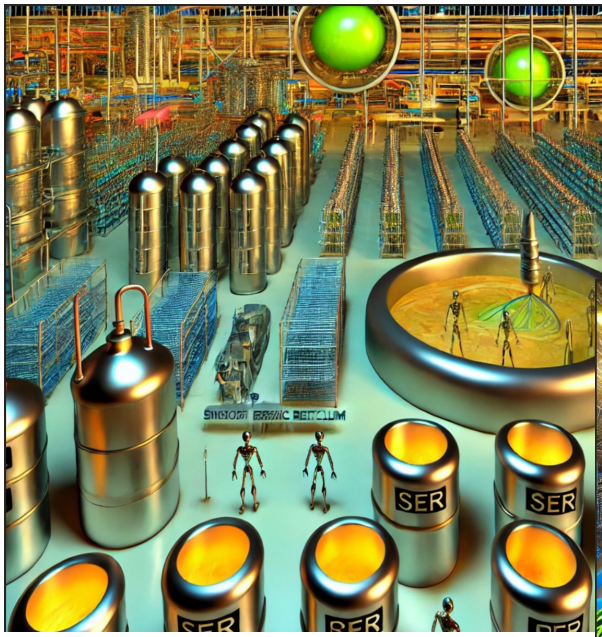
Endoplasmic Reticulum



- Some organelles process and sort proteins and exchange material with the environment.

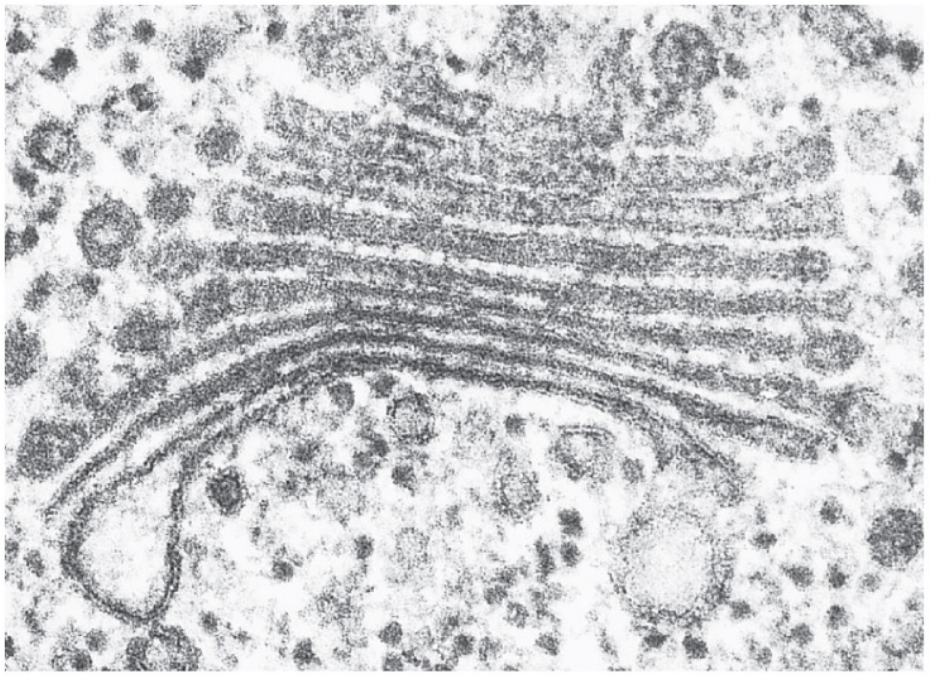
- The endoplasmic reticulum (ER) is a series of membranous sacs in the cytoplasm.

- There are two types of ER:
  - Rough ER has ribosomes associated with it and plays a role in protein processing.
  - Smooth ER lacks ribosomes and plays a variety of biochemical roles.



Golgi Complex

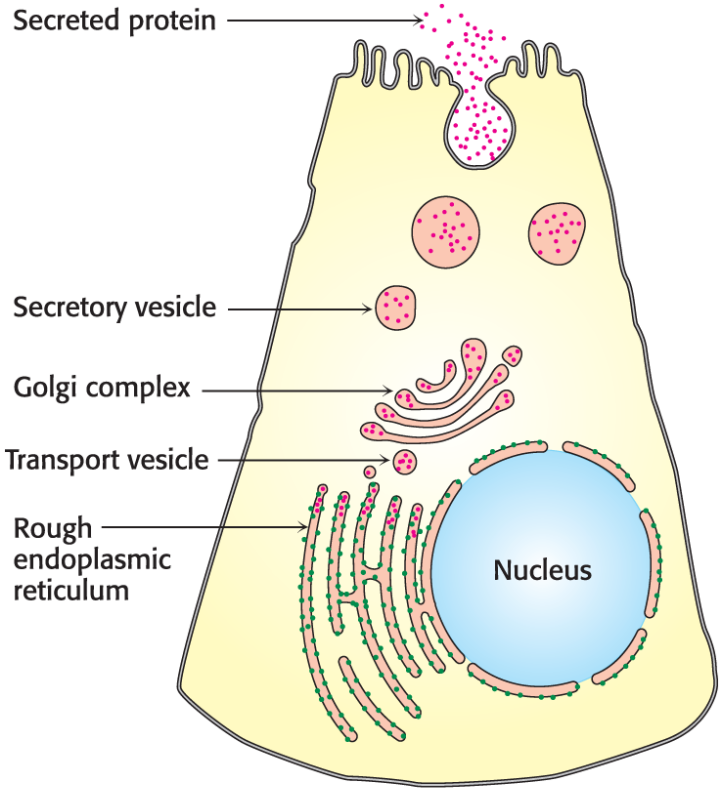
- The Golgi complex is a series of stacked membranes that play a role in protein sorting. Carbohydrates are also attached to proteins in the Golgi complex.
- Proteins are shuttled from the rough ER to the Golgi complex by transport vesicles.

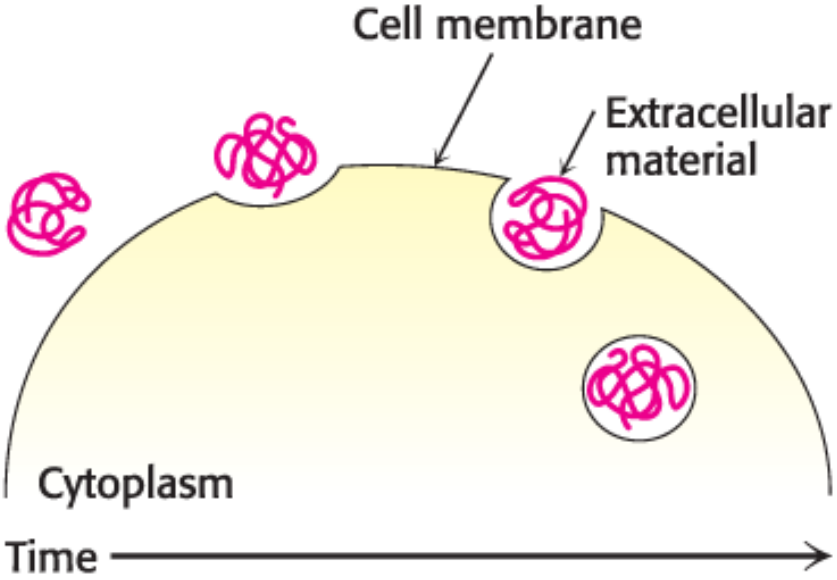


Courtesy of L. Andrew Staehelin, University of Colorado.

Exocytosis and Endocytosis

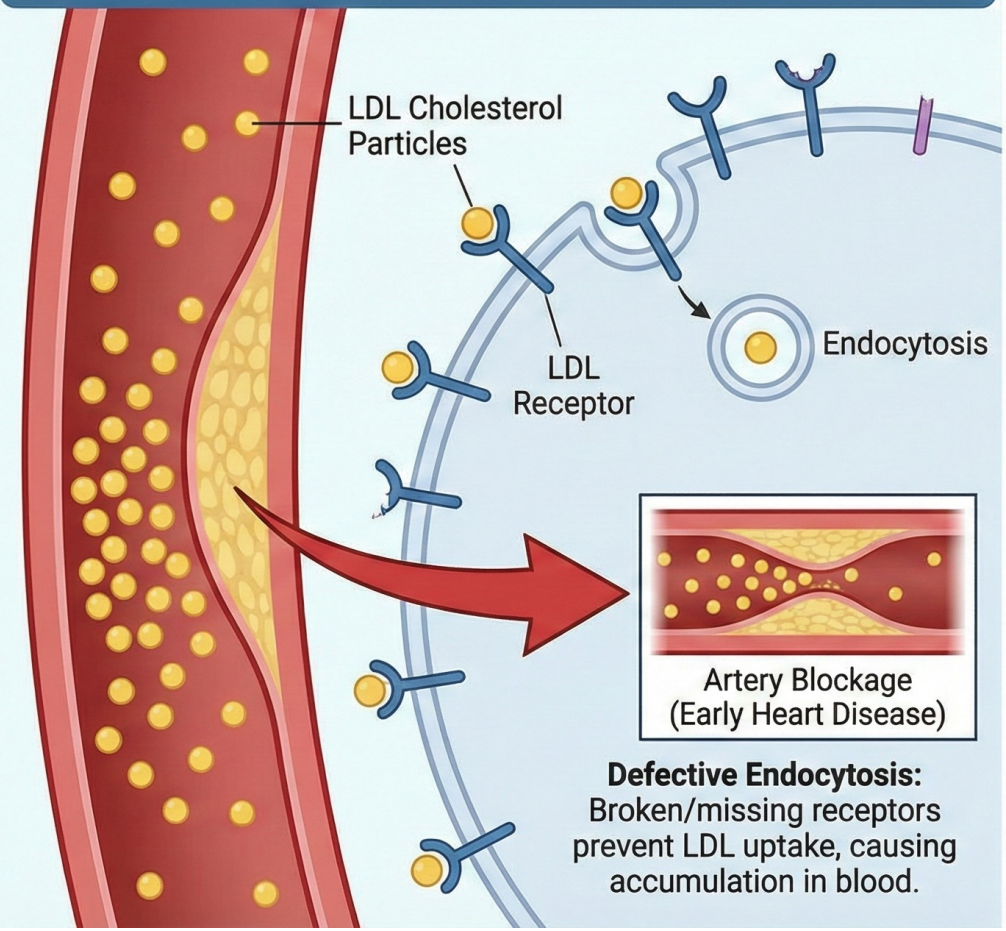
- Secretory granules secrete biomolecules in the process of exocytosis.
- Endocytosis is a means of bringing crucial biomolecules into the cell.
- An endosome is the structure that forms when the plasma membrane invaginates and buds off.





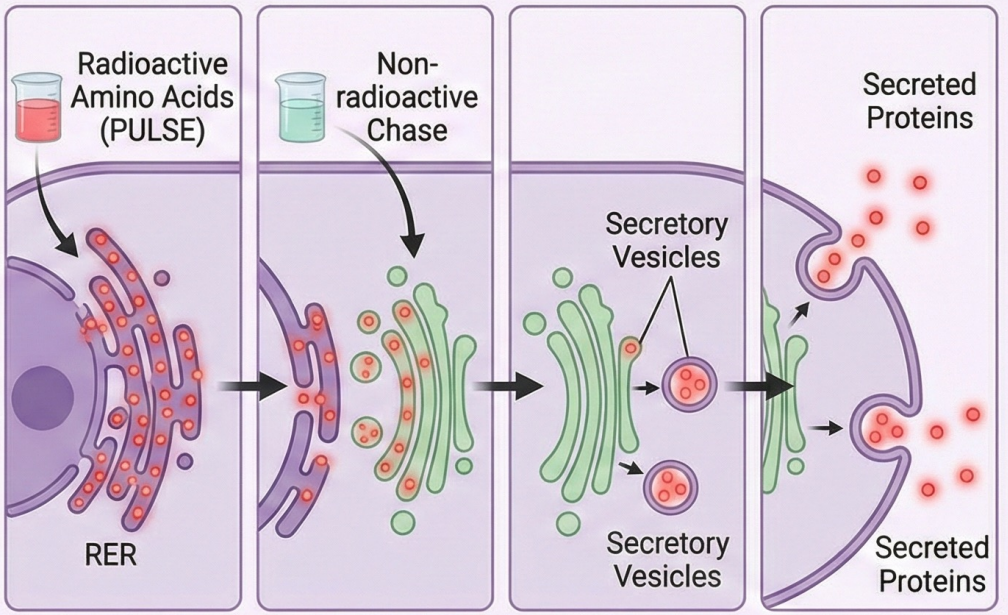
THE LEVEL UP: MEMBRANE DYNAMICS & PROTEIN TRAFFICKING

CLINICAL CORRELATE: FAMILIAL HYPERCHOLESTEROLÉMIA (FH)



Defective Endocytosis: Broken/missing receptors prevent LDL uptake, causing accumulation in blood.

RESEARCH EDGE: PULSE-CHASE EXPERIMENT (PROTEIN SECRETION)



PULSE: Newly made proteins in RER get labeled.

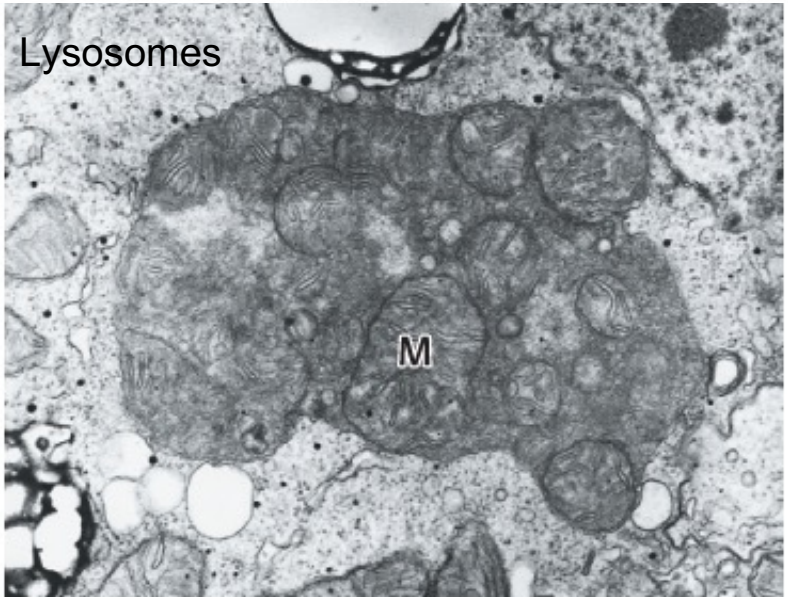
CHASE: Labeled proteins move to Golgi.

CHASE: Proteins packaged into secretory vesicles.

SECRETION: Labeled proteins are released.

Phagocytosis and Lysosomes

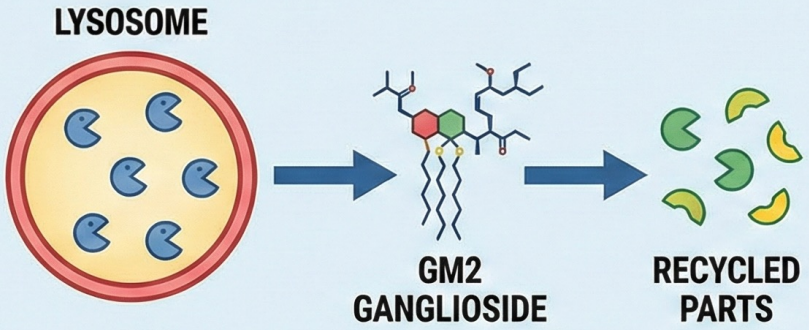
- Large amounts of material can be taken into the cell by the process of phagocytosis.
- Lysosomes contain a variety of digestive enzymes.
- Lysosomes fuse with endosomes to digest material brought into the cell.



TAY-SACHS DISEASE: LYSOSOMAL DYSFUNCTION & CLINICAL IMPACT

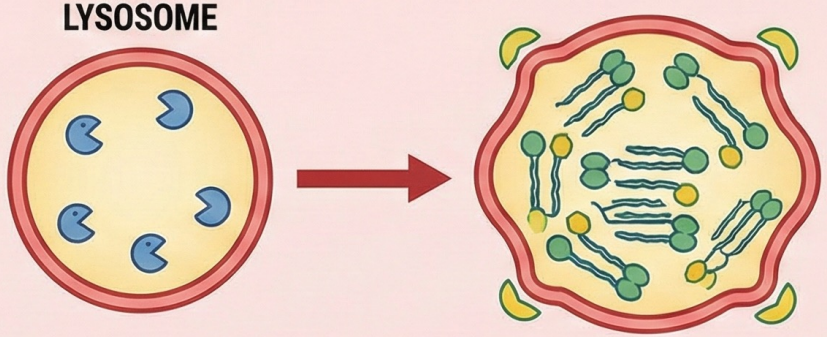
MECHANISM: THE BROKEN RECYCLING CENTER

HEALTHY NEURON



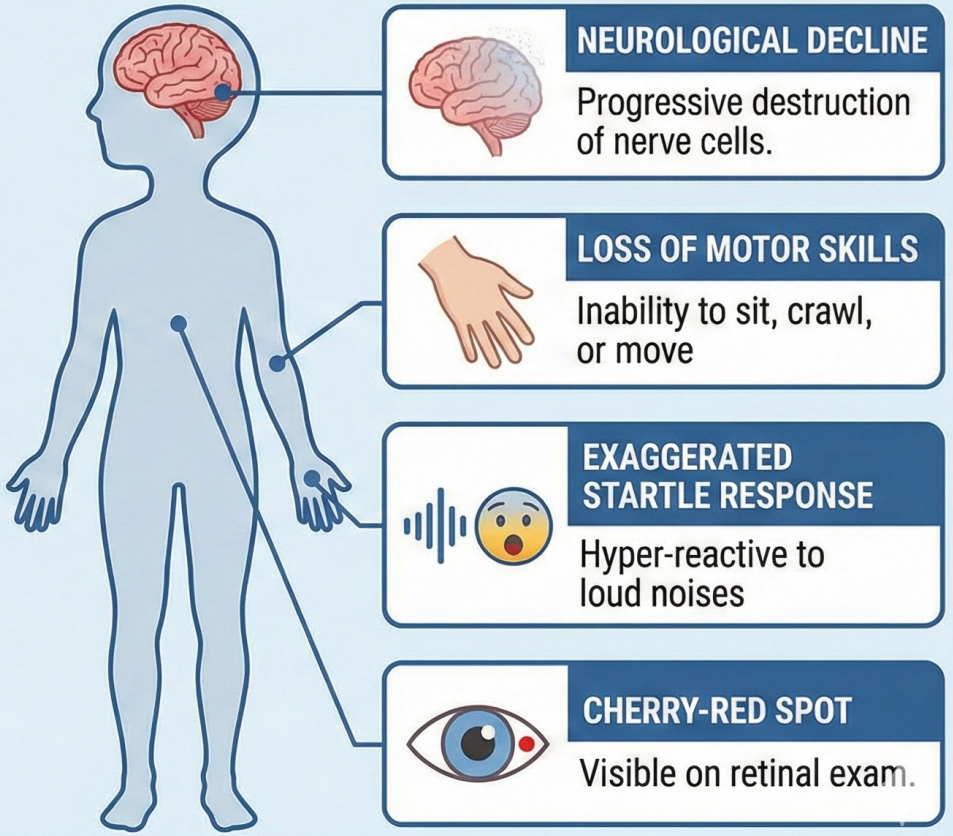
Functional Hex A Enzyme breaks down GM2 Ganglioside.

TAY-SACHS NEURON



MISSING Hex A Enzyme: GM2 accumulates, damaging cell.

CLINICAL IMPLICATIONS



NEUROLOGICAL DECLINE
Progressive destruction of nerve cells.

LOSS OF MOTOR SKILLS
Inability to sit, crawl, or move

EXAGGERATED STARTLE RESPONSE
Hyper-reactive to loud noises

CHERRY-RED SPOT
Visible on retinal exam.